

LOGISTICS AT THE URBAN EDGE

Bachelor Diploma Project

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- 01** **STUDIO BRIEF**
Abstract, types of logistics centers
- 02** **PROBLEM STATEMENT**
Main topic
- 03** **SITE ANALYSIS**
Information and research about the location
- 04** **CASE STUDY**
Analysis of an existing warehouse
- 05** **URBAN SCENARIOS**
Mobility, buffer, zoning
- 06** **DESIGN RESPONSE**
Proposed masterplan, spatial experience
- 07** **DISCUSSION OF RESULTS**
Findings
- 08** **DIPLOMA REPORT**
Technical report
- 09** **EPILOGUE**
Project conclusion

STUDIO BRIEF

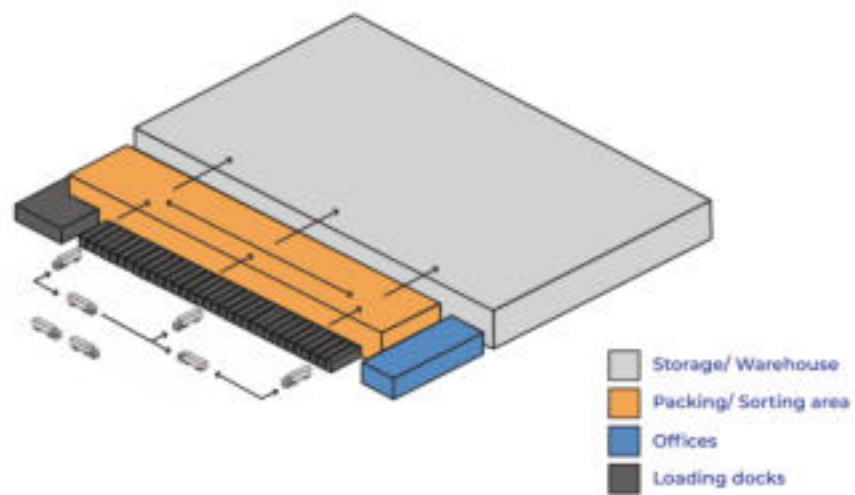
The studio focuses on **rethinking logistics architecture** as an active component of the urban and environmental system rather than a purely technical infrastructure. In collaboration with industry partners such as Ramboll, Henning Larsen, and Panattoni, the work engages with real-world challenges related to the design and performance of logistics facilities.

Emphasis is placed on evidence-based design, using measurable criteria such as energy, carbon, biodiversity to inform decision-making. Through a **combination of analytical research and speculative design**, the work explores how logistics buildings can evolve from mono-functional objects into adaptable, context-responsive systems with broader environmental and urban value.



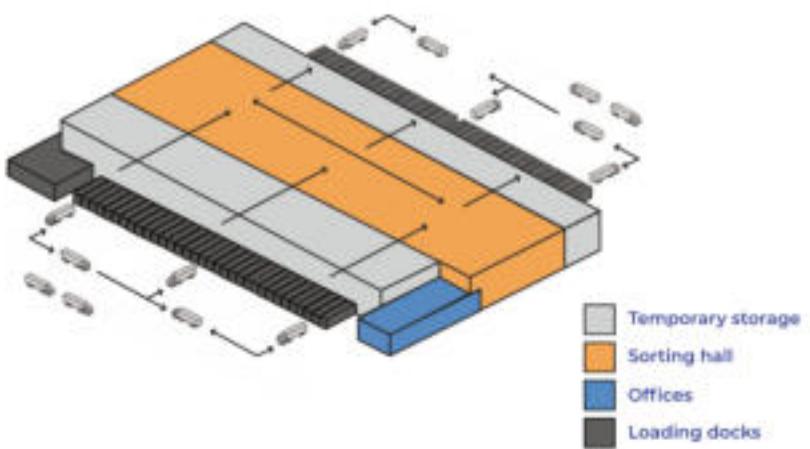
TYPES OF LOGISTICS CENTERS

BIG-BOX



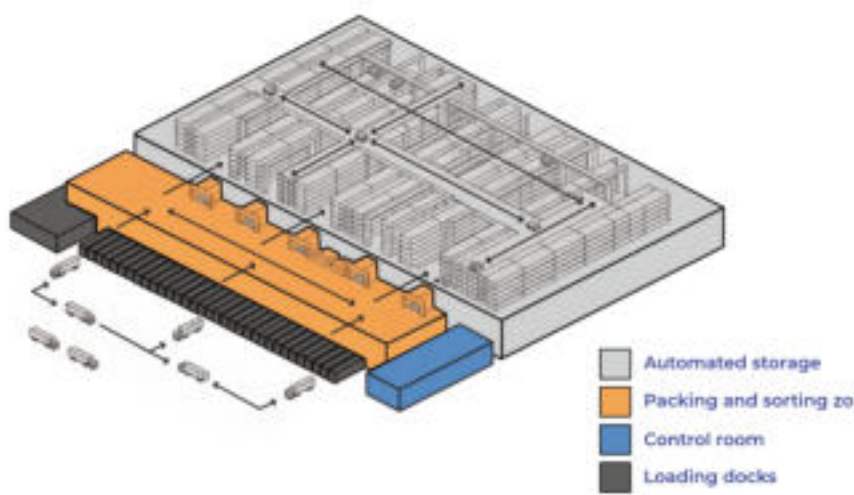
Big-box logistics buildings are very large warehouses used for storing and distributing goods to shops or customers. They usually have a simple rectangular shape with many loading docks on one side. These buildings are often located near highways so trucks can easily access them.

CROSS-DOCK



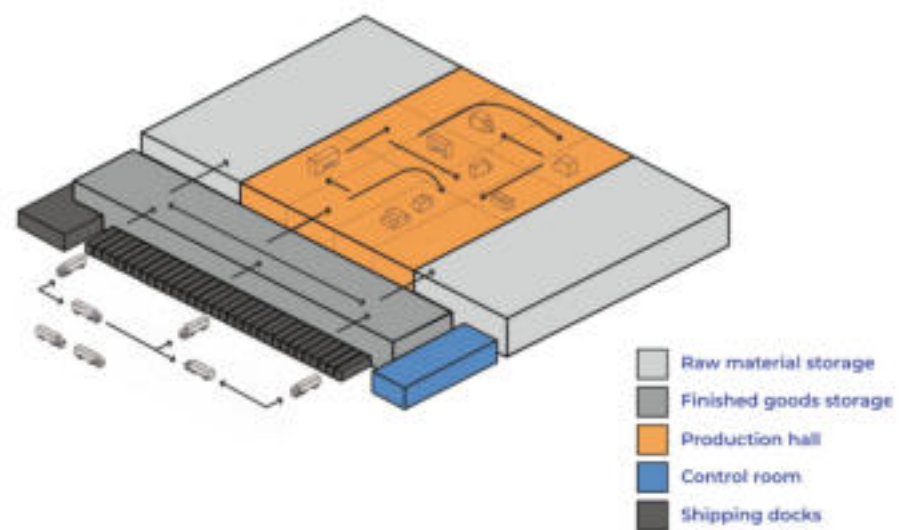
Cross-dock facilities are designed for very fast distribution. Instead of storing goods, they quickly transfer them from incoming vehicles to outgoing vehicles. These buildings are often located near airports or major highways.

AUTOMATED WAREHOUSE



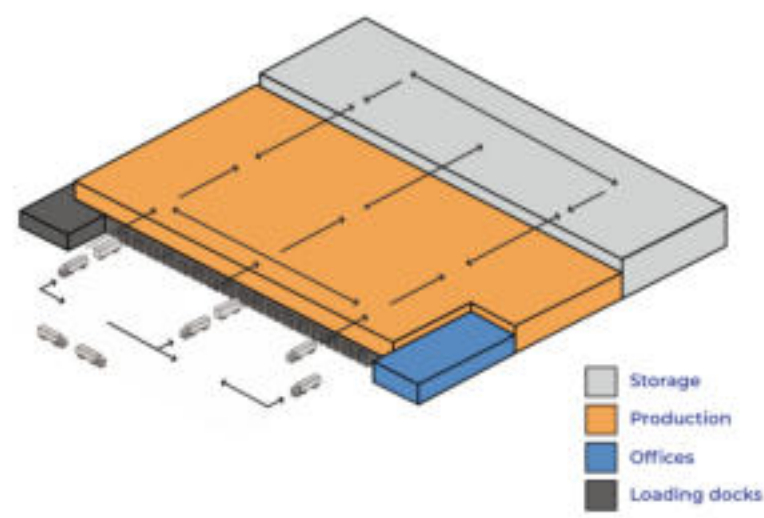
Automated warehouses use advanced technology such as robots, conveyors, and automated storage systems. These buildings can process large numbers of orders very quickly.

INDUSTRIAL MANUFACTURING



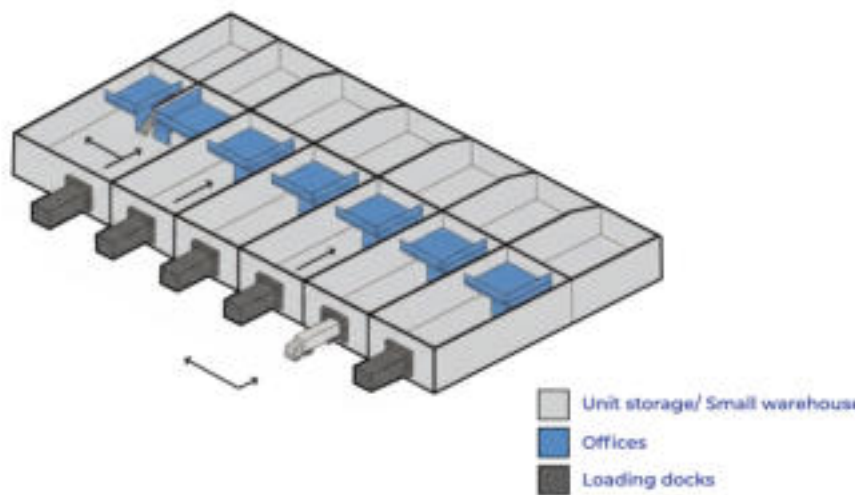
These logistics facilities support large factories. They store parts and materials that are needed for production and also handle the distribution of finished products.

LIGHT INDUSTRIAL



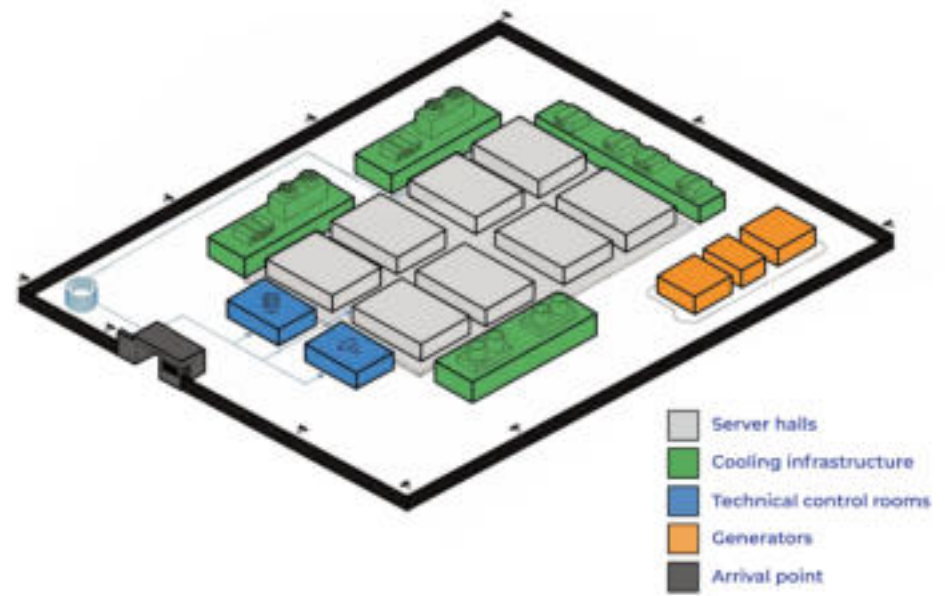
This type combines light manufacturing with logistics. Some products may be assembled, repaired, or prepared before being shipped. These facilities are usually located in industrial zones near cities.

MULTI-TENANT



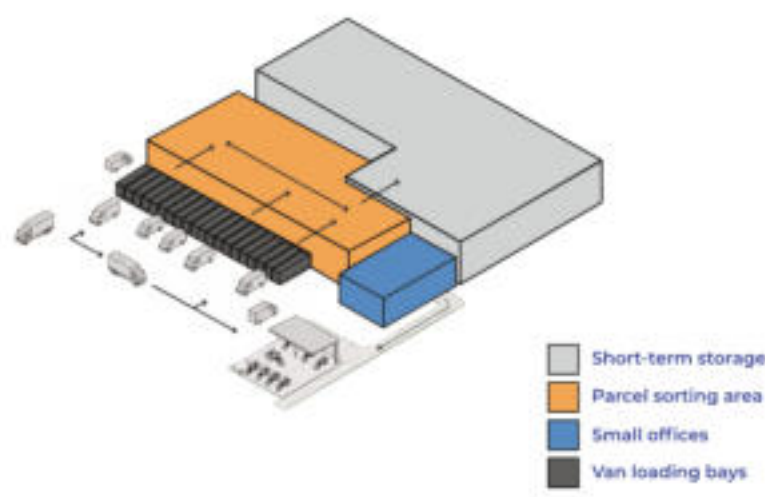
A multi-tenant business park is a large logistics area with several warehouses used by different companies. All buildings share the same infrastructure such as roads, parking, and truck circulation.

DATA CENTER



Data centers store and process digital information instead of physical goods. They contain large server rooms and complex cooling systems. These buildings require reliable electricity and strong internet connections.

LAST-MILE

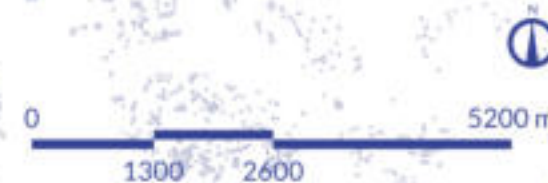


Urban logistics centers are smaller facilities located inside cities. They help deliver packages quickly to nearby neighborhoods.

PROBLEM STATEMENT

The Prague metropolitan region contains **over 3.2 million m² of modern logistics and industrial space**, positioning it as a key distribution hub within Central Europe. This concentration reflects broader patterns of rapid logistics expansion driven by global supply chains, e-commerce growth, and infrastructural accessibility. However, the scale and speed of this development have also produced significant spatial and environmental challenges, particularly at the urban periphery where large mono-functional facilities dominate.

The problem addressed by the project is the **current separation between logistics efficiency and urban-environmental quality**: while logistics zones are highly optimized for storage, transport, and economic performance, they often create fragmented landscapes, impermeable surfaces, weak public accessibility, ecological disruption, and tensions with nearby residential areas. The project therefore asks how **logistics architecture can be rethought as an active part of the urban and environmental system**, rather than as an isolated technical infrastructure.



~3.2 million m² of of logistics and industrial space

SITE ANALYSIS

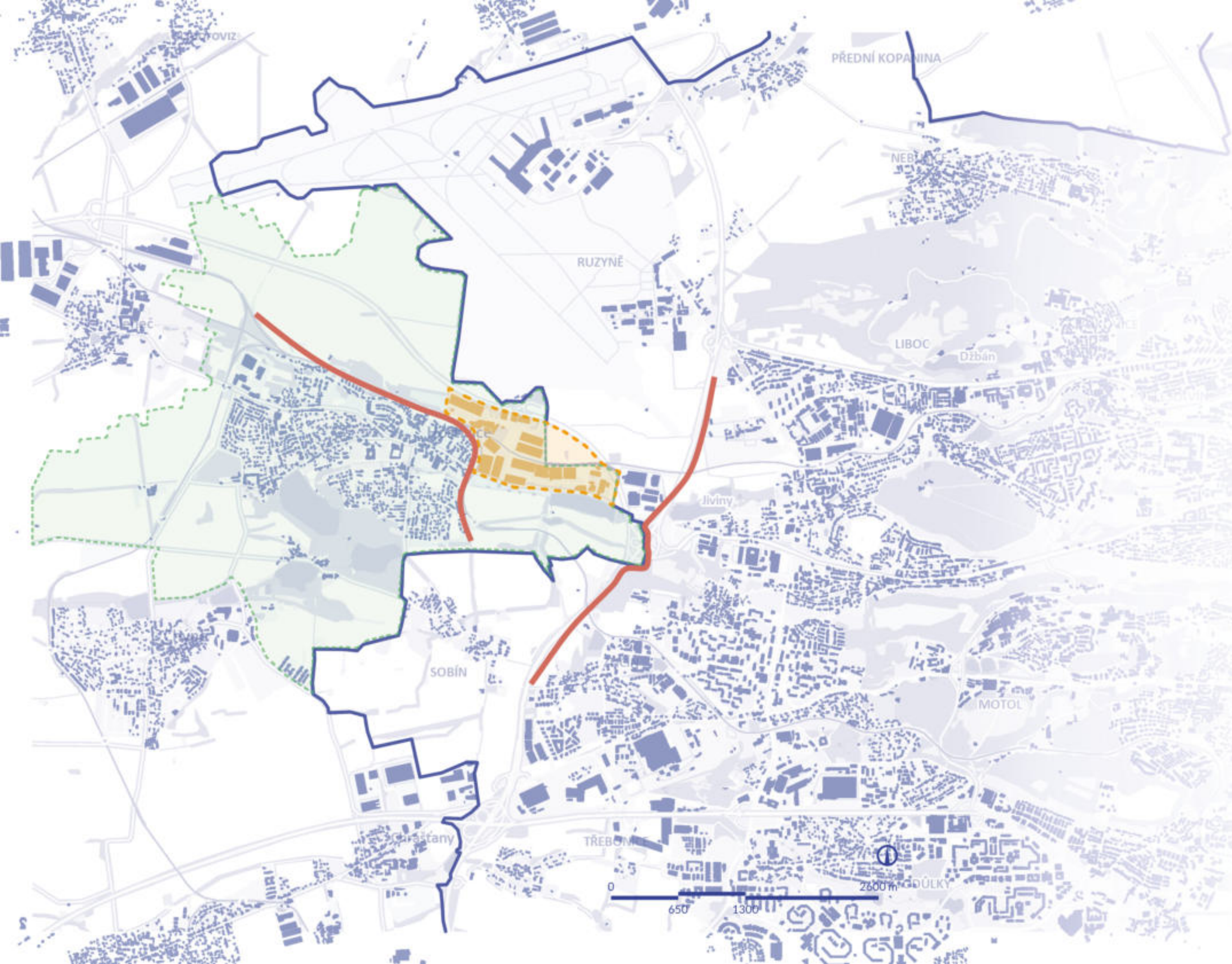
The project is located in the **Hostivice industrial zone** on the western edge of Prague. The area, with a perimeter of approximately **4,700 m** and a total surface of around **876,000 m²**, has developed quickly over the last three decades, mainly because of its strong connection to infrastructure. It is close to major roads, railway lines, and Václav Havel Airport Prague, which makes it attractive for logistics development and connects it to wider regional and international networks.

Despite its strategic accessibility, the industrial zone exists in close **contact with pre-existing residential neighborhoods**, creating spatial and social tensions. The coexistence of large-scale logistics operations and everyday living environments raises questions about **land-use compatibility** and the long-term impact of such developments on surrounding communities.

Large-scale logistics buildings dominate the area, often forming **fragmented urban conditions** with limited walkability, and public accessibility. Their scale and layout frequently prioritize efficiency and vehicular movement over human-scale experience. This context provides a relevant case for examining how contemporary logistics architecture interacts with its surroundings and how it might be rethought to better respond to long-term urban and environmental challenges.







STRENGTHS

The site benefits from a strategic location with strong connections to Prague and regional infrastructure, forming part of an established and economically active logistics ecosystem, while its large plot sizes allow flexibility for spatial reconfiguration, the terrain supports easy construction and adaptation, and its proximity to surrounding landscape systems offers potential links to green and water networks.

WEAKNESS

The area suffers from monofunctional zoning that limits diversity and urban life, is dominated by large-scale warehouses with poor human scale and strong visual impact, and creates a barrier effect with limited permeability for pedestrians and cyclists, further intensified by traffic conflicts between heavy trucks and local mobility, low environmental performance including heat island effects, runoff, and limited biodiversity, and a weak public realm lacking meaningful shared or civic spaces.

OPPORTUNITIES

There are clear opportunities to reconnect fragmented ecology through green corridors and water systems, introduce a programmatic mix combining work, community, and landscape, improve mobility networks by integrating cycling and pedestrian infrastructure, transform rigid edges into active and accessible interfaces, and apply data-driven design through KPIs to achieve measurable environmental improvements.

THREATS

These potentials must be balanced against threats such as developer-driven efficiency pushing toward standard big-box solutions, regulatory constraints related to zoning and safety requirements, increasing traffic demand, ongoing noise and pollution conflicts with nearby housing, economic pressures that challenge sustainable solutions, and the risk of greenwashing through superficial environmental gestures without real impact.

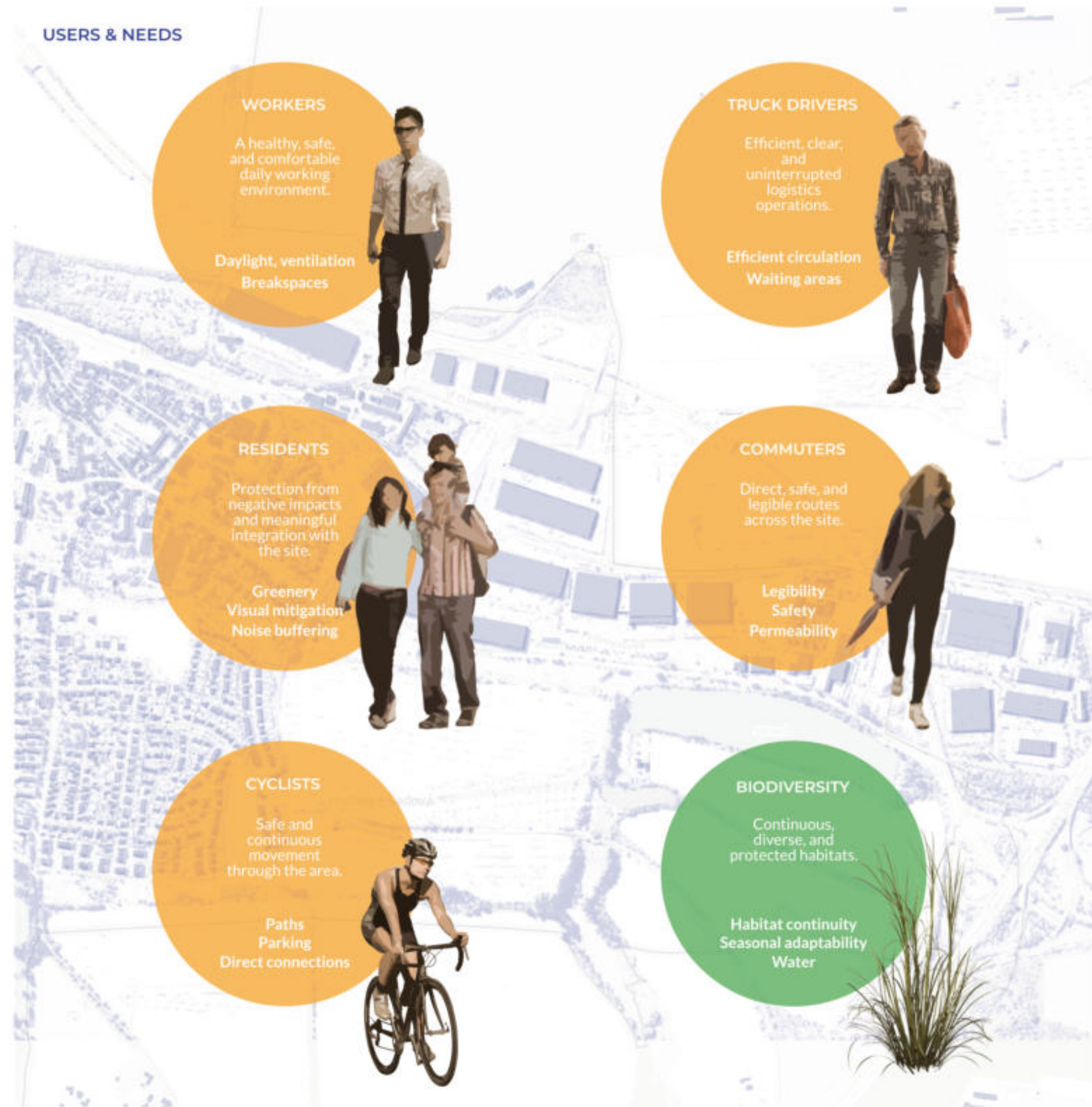
1988-1989



CURRENT SITUATION







CASE STUDY

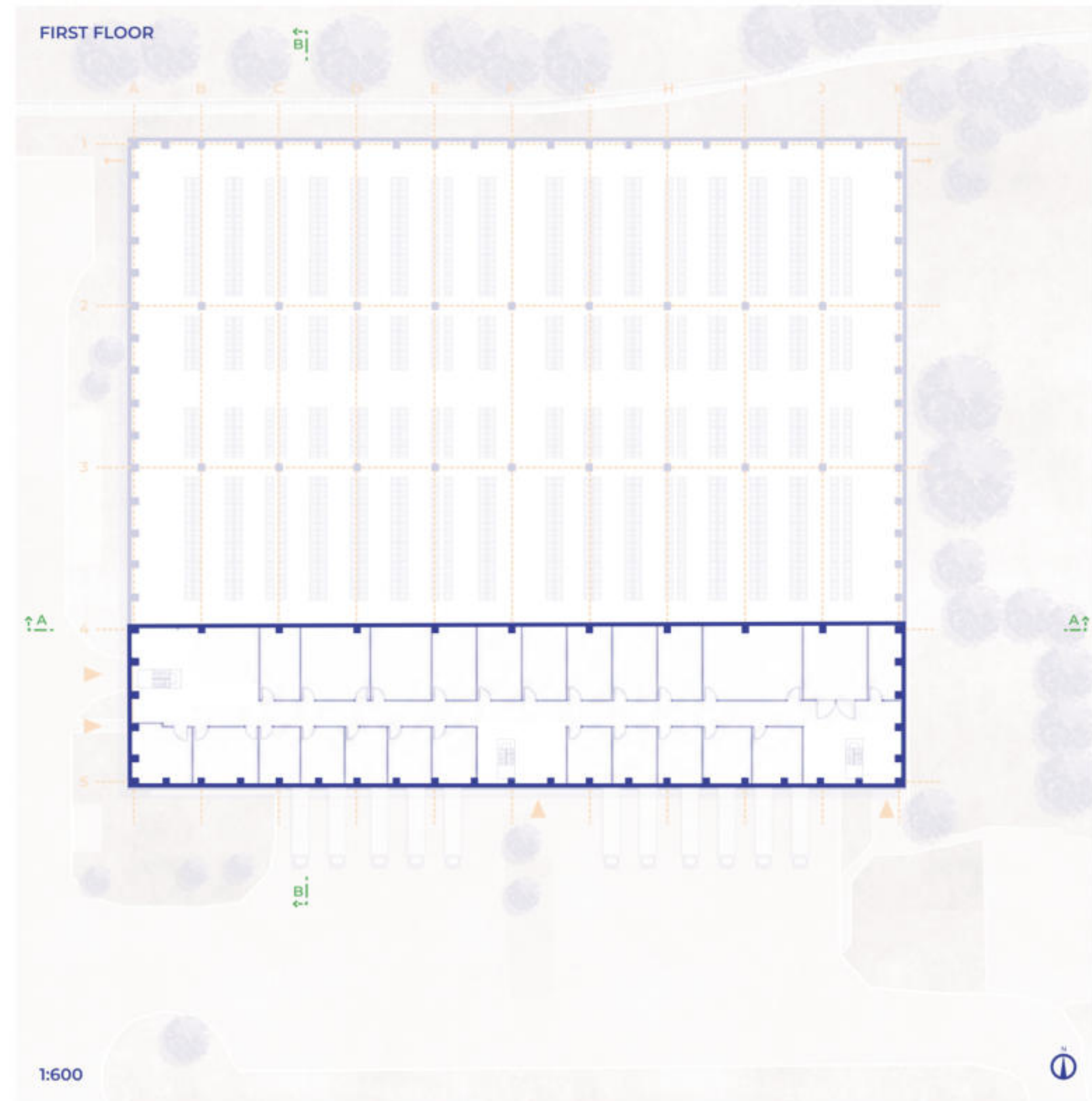
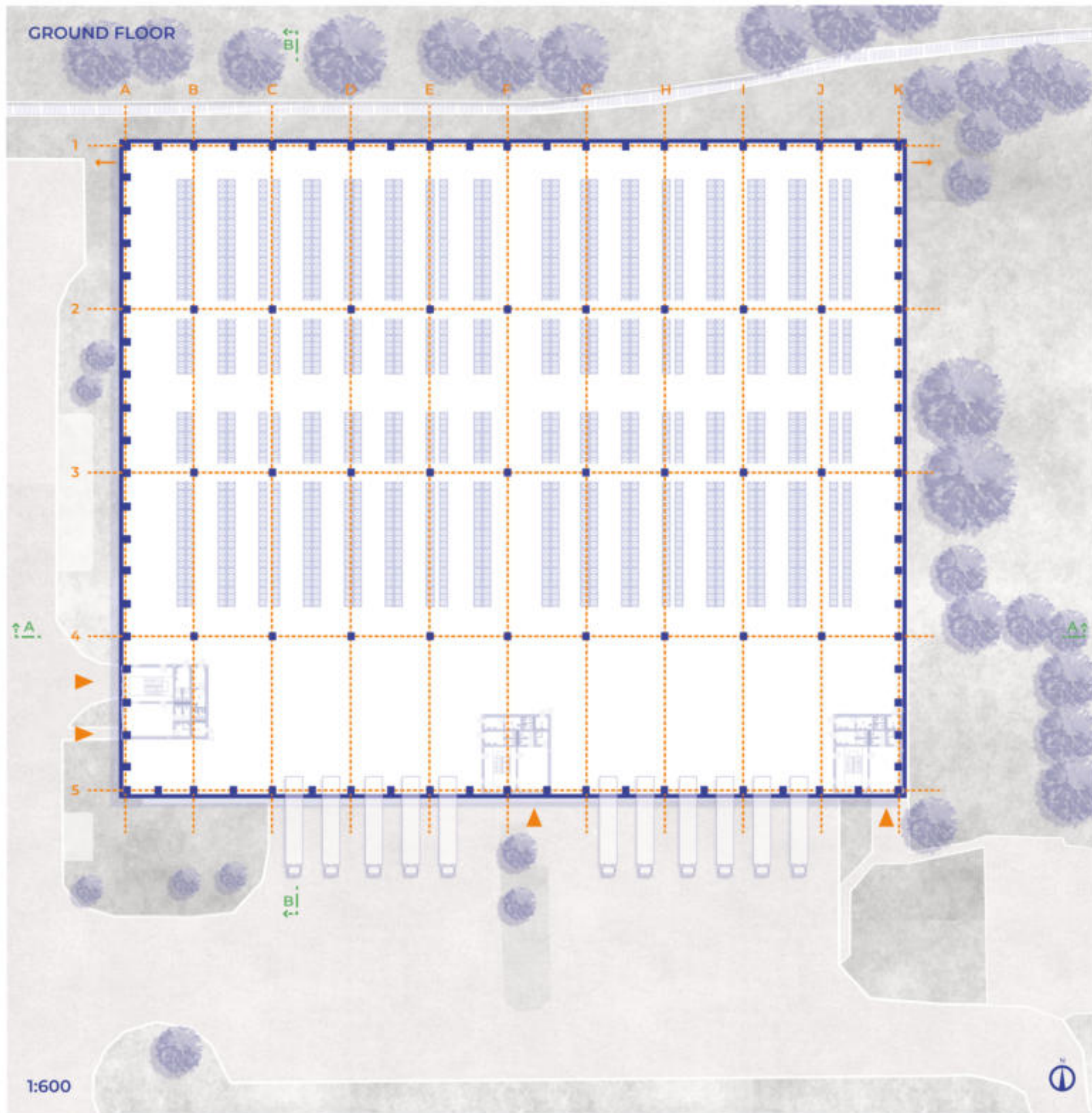
Building on the project context, the research moves from the scale of the industrial zone to the scale of a single building. The case study **focuses on one existing warehouse within the Hostivice industrial zone** in order to understand how this type of logistics building works in more detail.

The logistics buildings on the site are highly **standardized**. Although they differ in size, they follow a similar typology, with comparable structural grids, circulation patterns, and functional layouts. Because of this repetition, one warehouse can be studied as a **representative example** of the wider logistics environment.

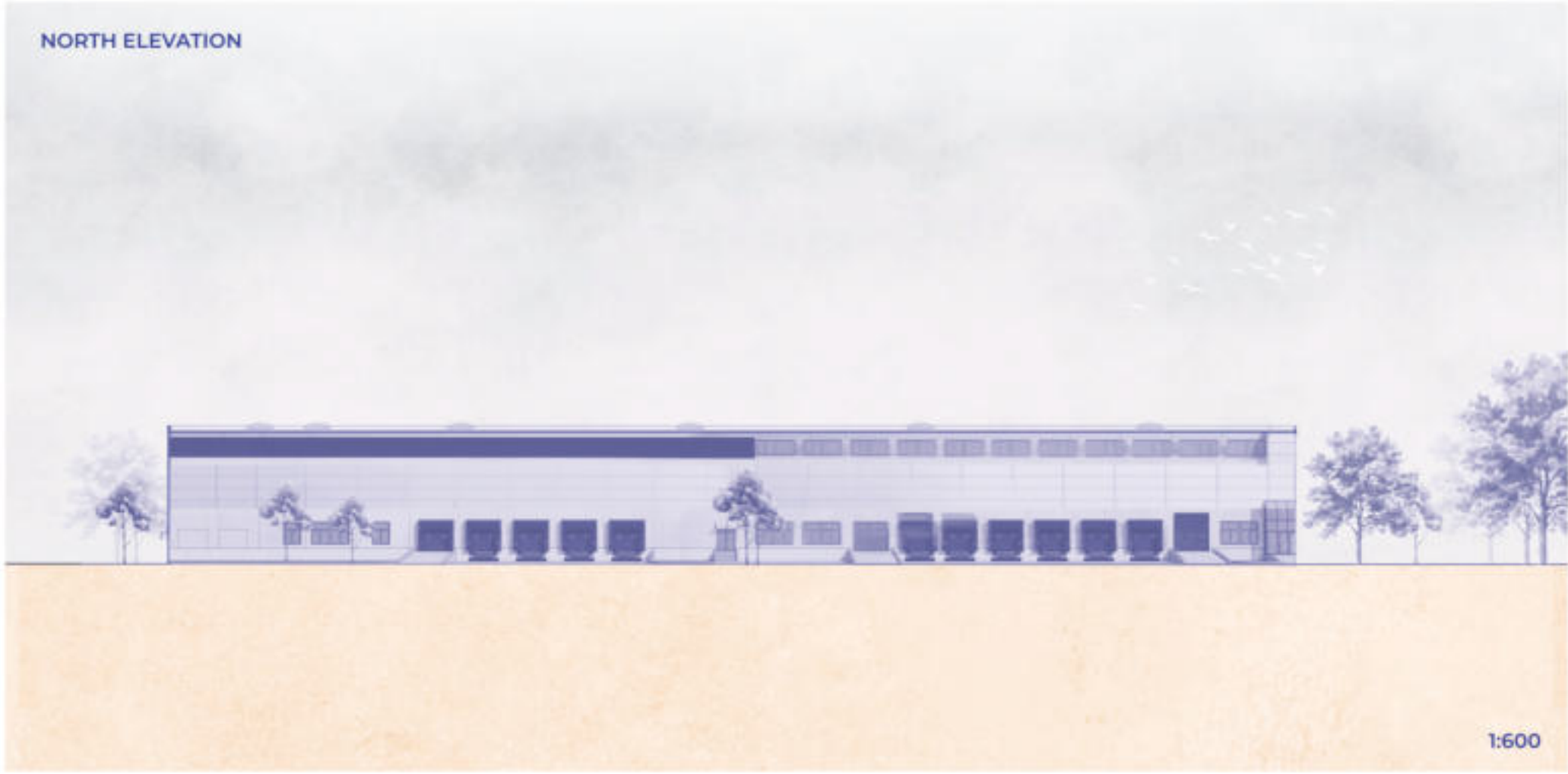
This analysis helps explain its spatial organization, construction system, and operational logic. By understanding the building at a deeper level, the case study creates a foundation for critically looking at the **current model of logistics architecture**.



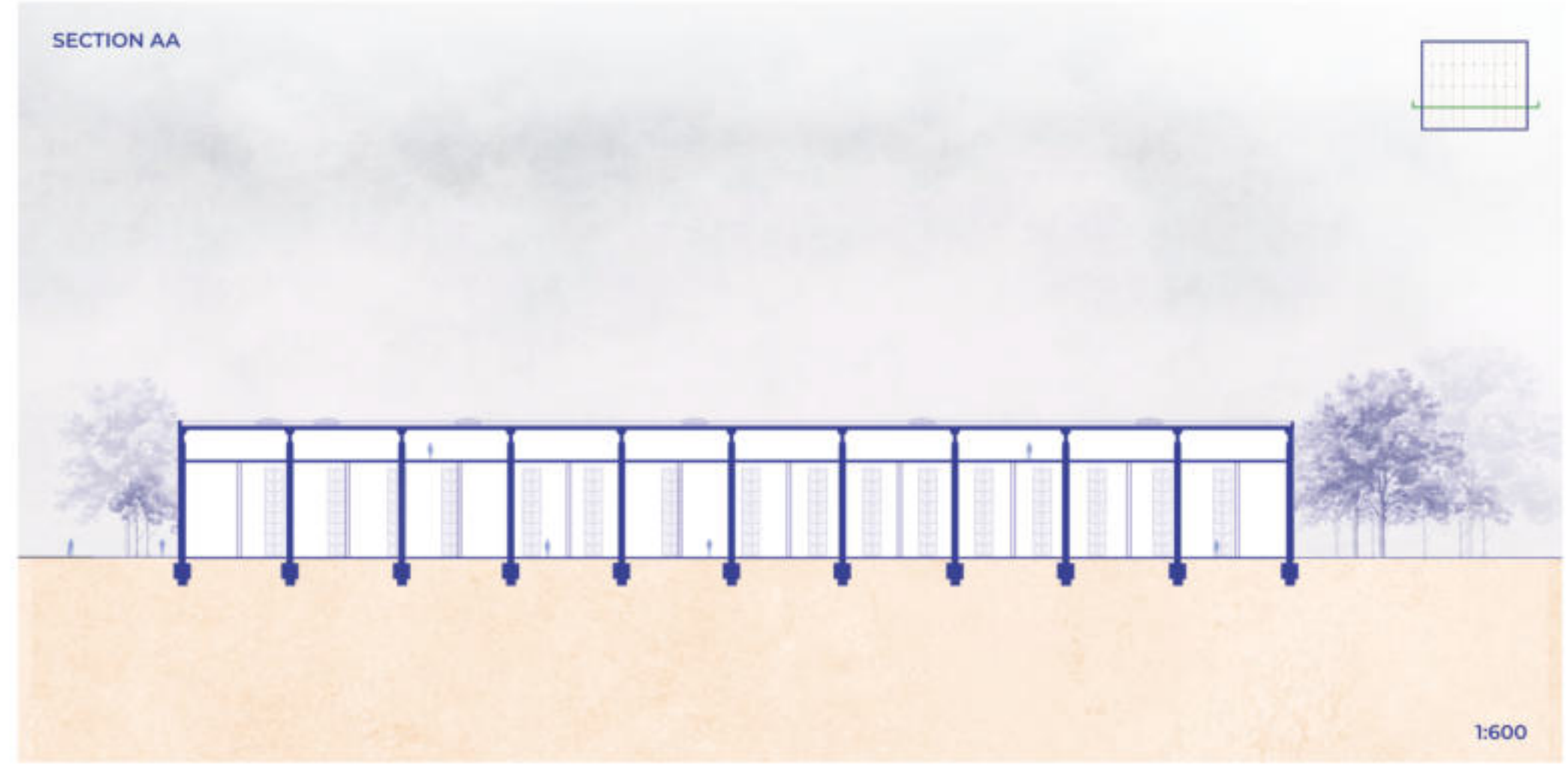




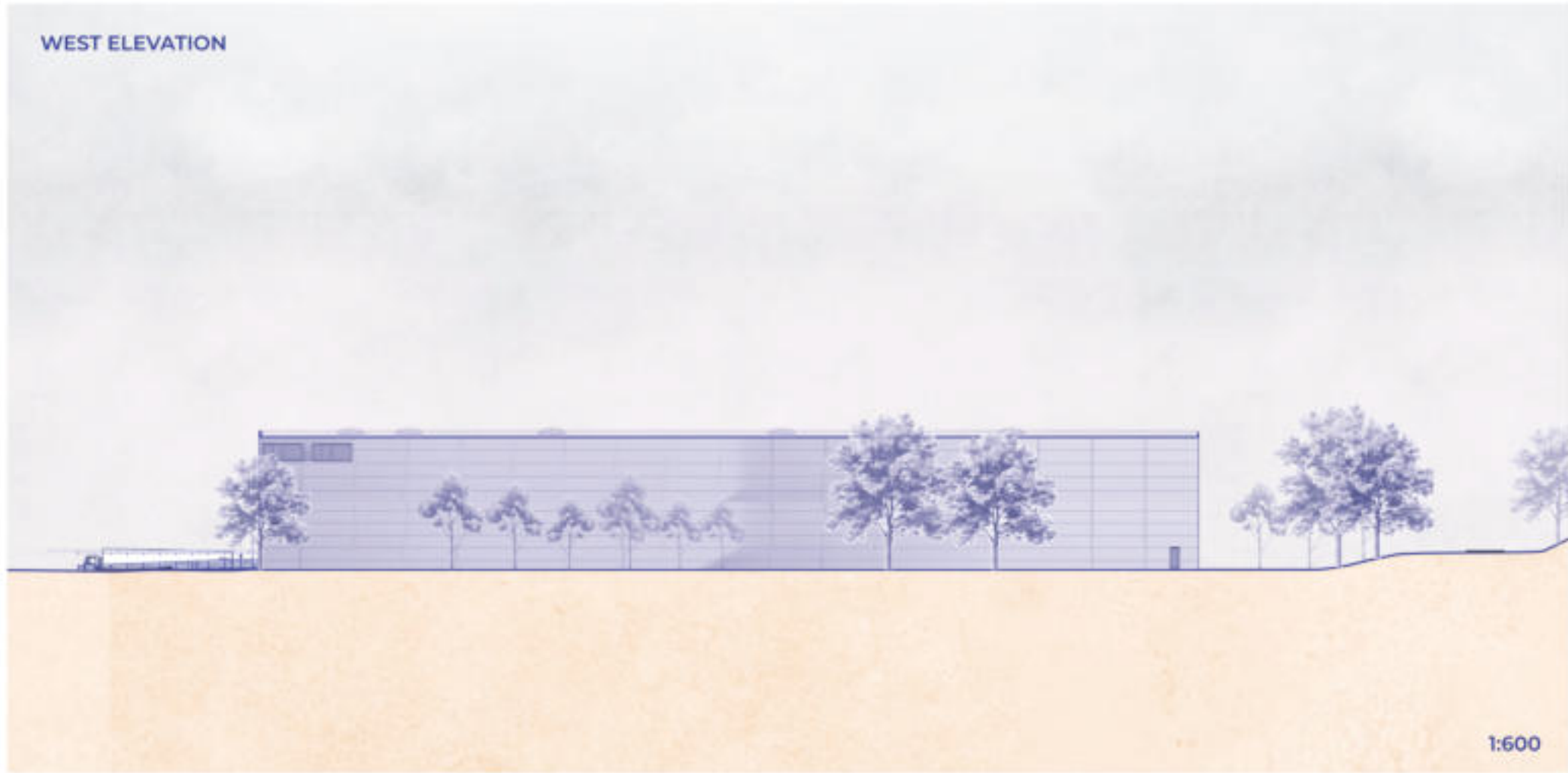
NORTH ELEVATION



SECTION AA



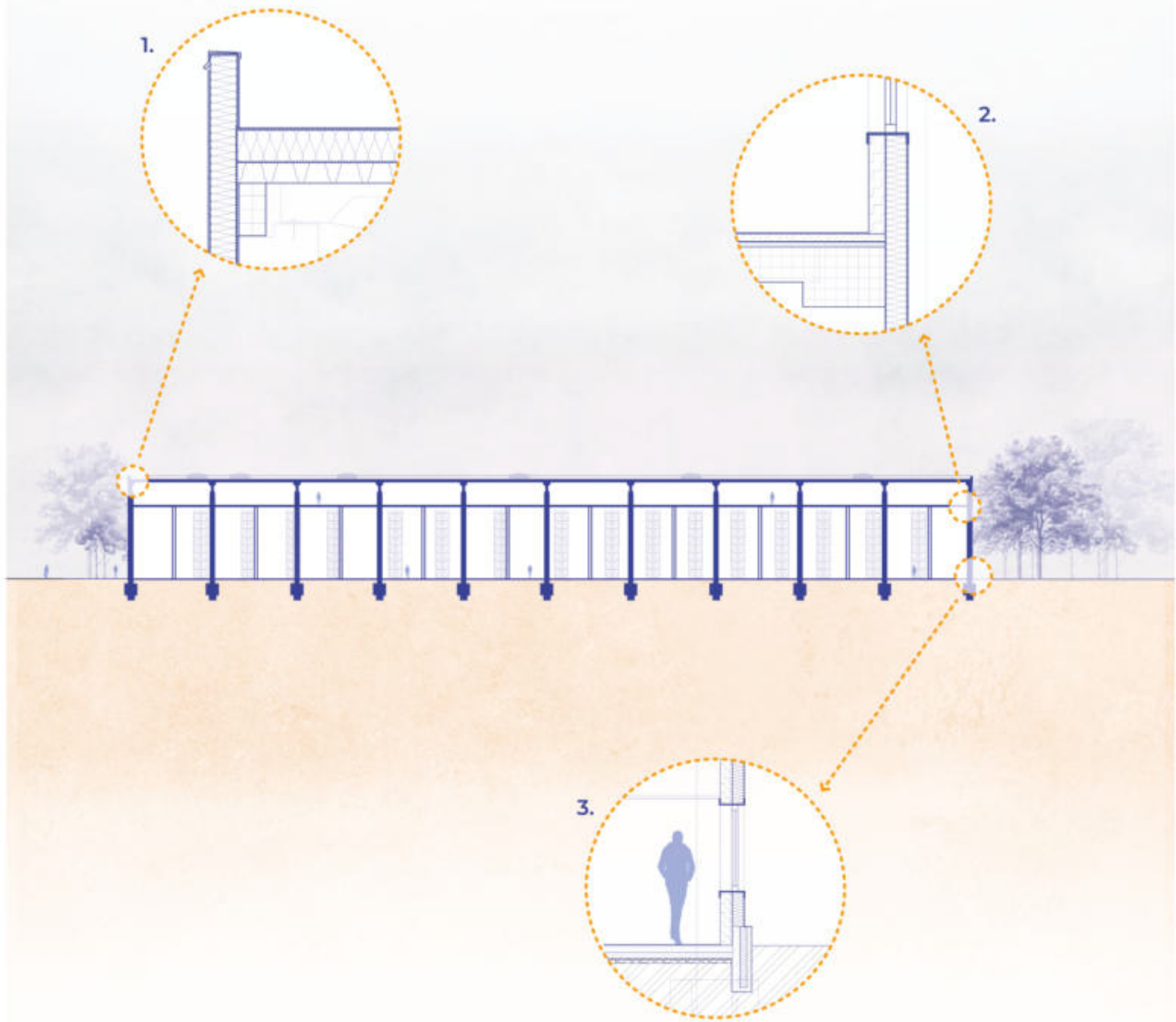
WEST ELEVATION



SECTION BB



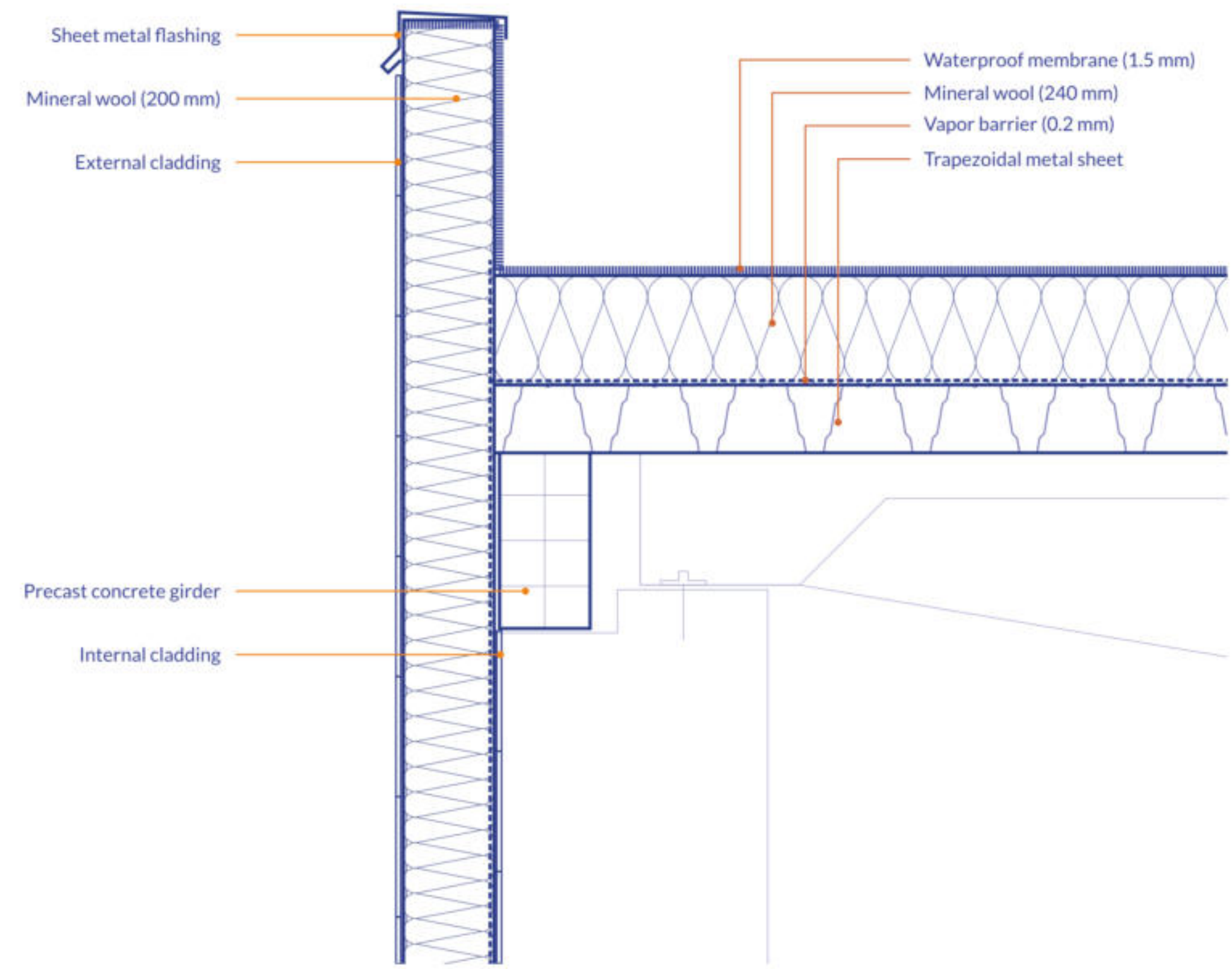
TECHNICAL DETAILS



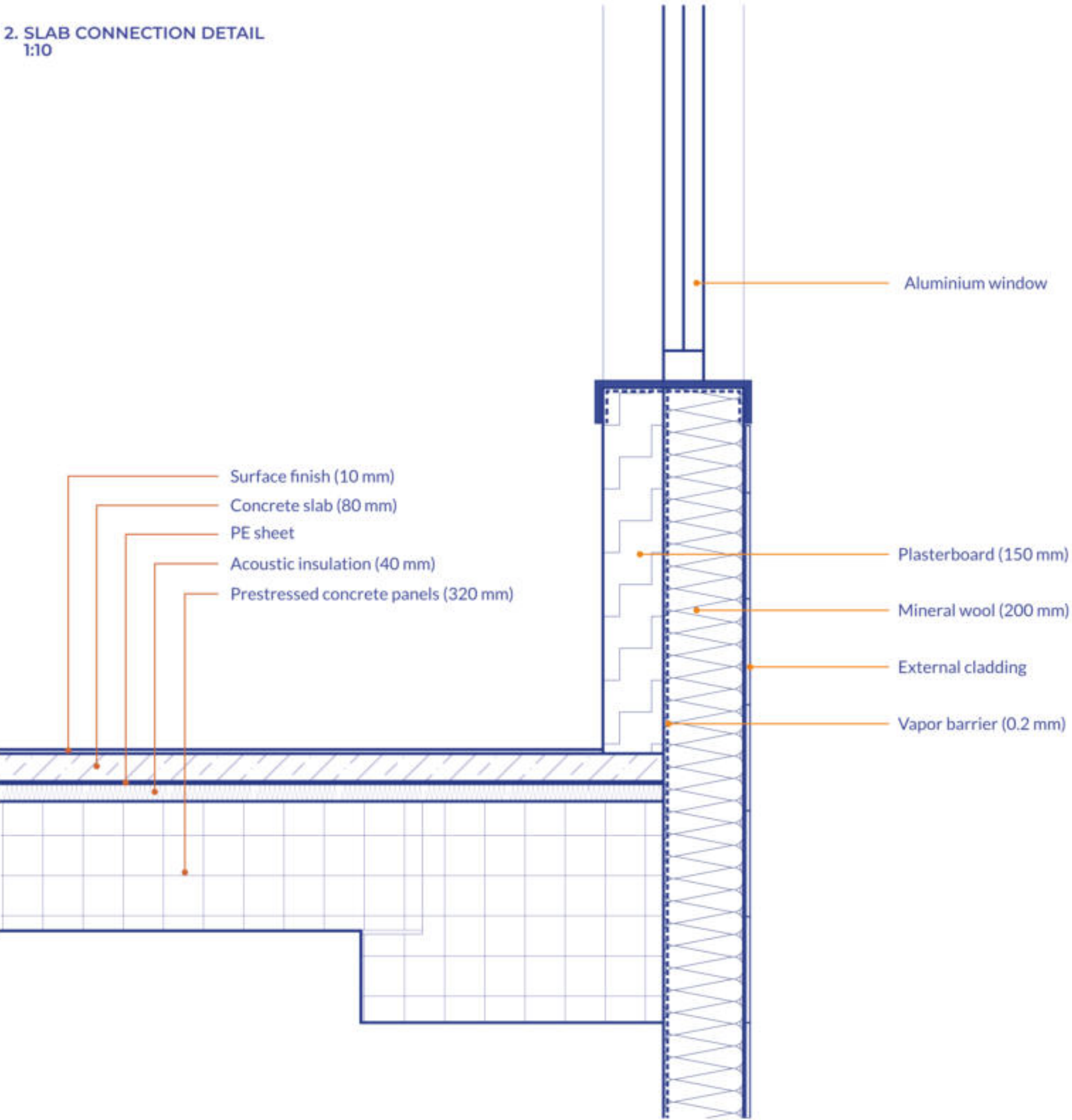
SECTION AA

1:600

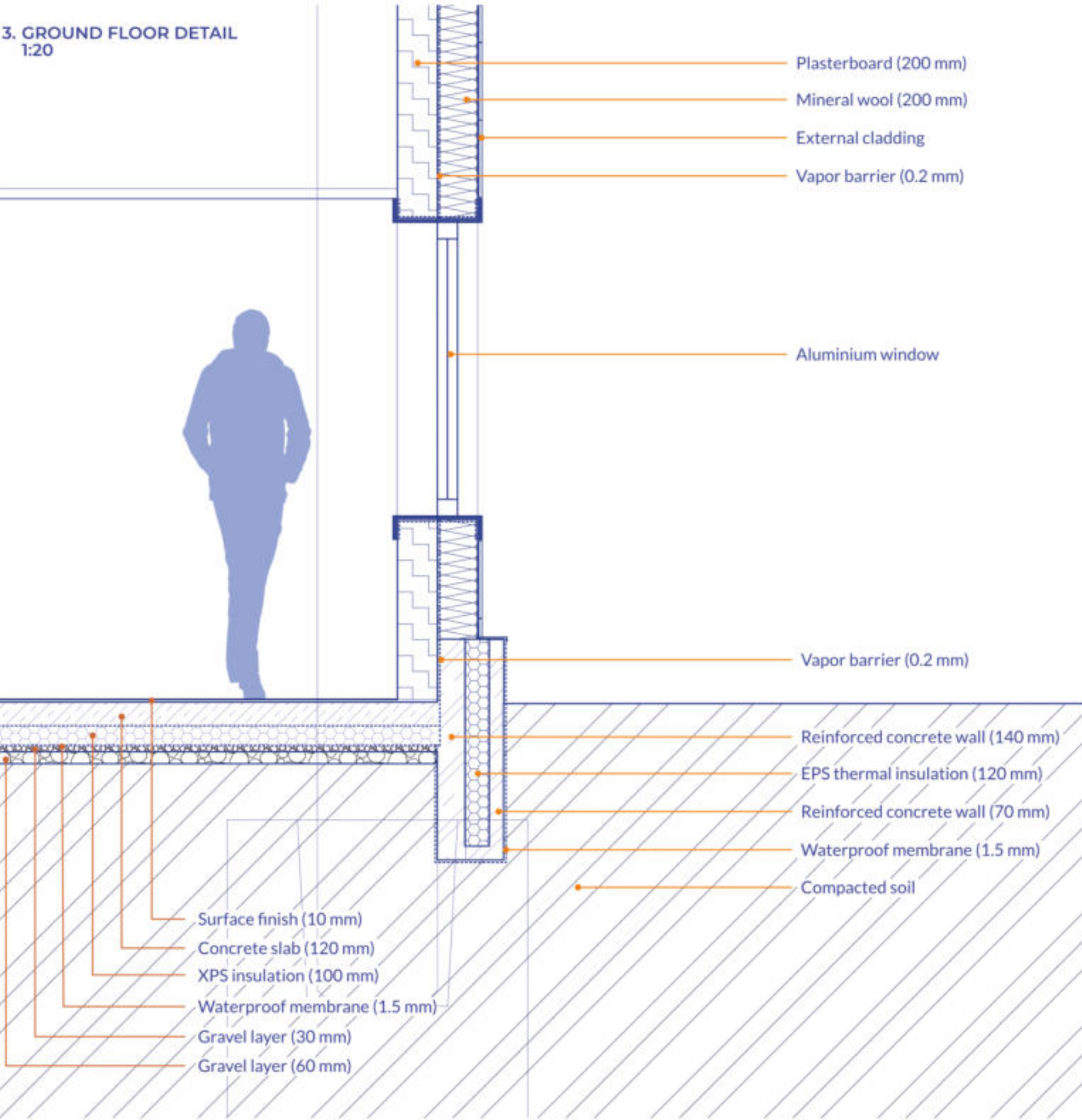
1. ROOF DETAIL
1:10



2. SLAB CONNECTION DETAIL
1:10



3. GROUND FLOOR DETAIL
1:20



URBAN SCENARIOS

The urban scenarios explore possible **ways of rethinking the relationship** between the **residential area** and the **industrial zone**. The aim is not to propose one fixed solution immediately, but to **test different directions** for how the two areas could coexist in the future.

Each scenario focuses on one specific aspect of the site. **Mobility** looks at how the connection between the two areas could be improved. **Green buffer zone** explores how greenery could create a softer separation, while improving visual and acoustic insulation. **Zoning** considers future development pressure from Prague and how this could influence the growth of the residential area and the evolution of logistics.

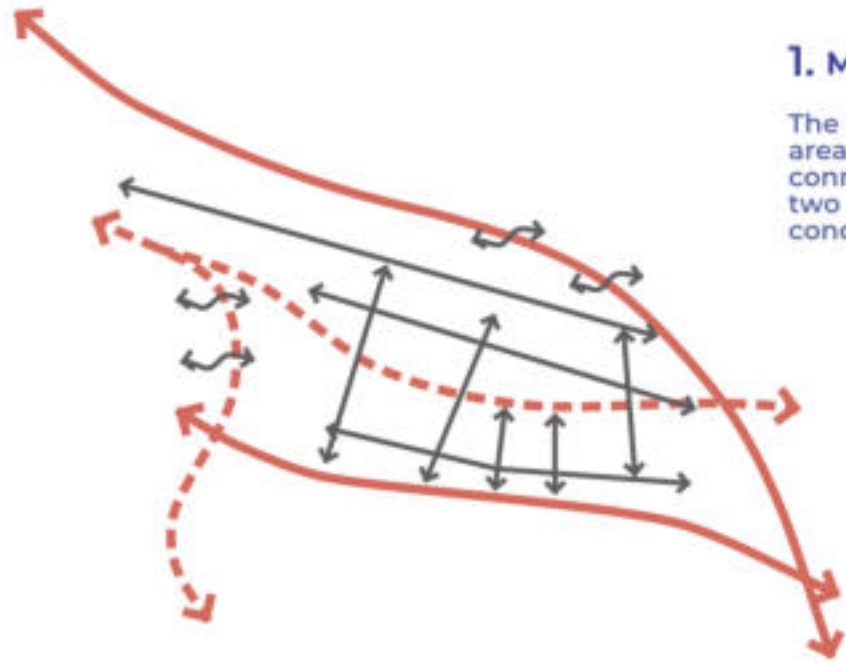
Together, these scenarios help clarify the **main urban challenges** of the site and create a framework for the next phase of the project.



SCENARIOS

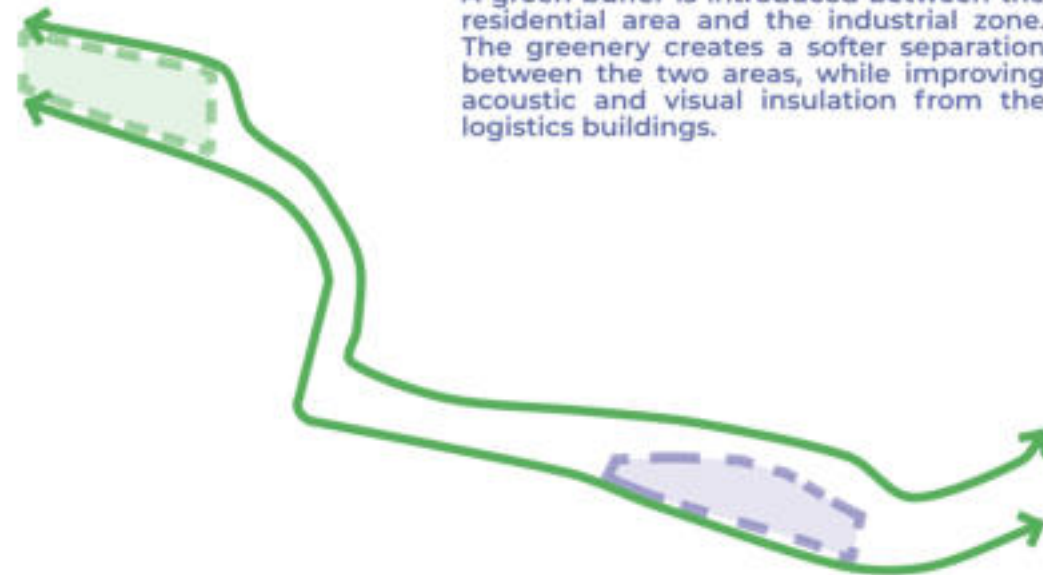
1. MOBILITY

The scenario looks at how the residential area and the industrial zone can be better connected. It proposes links between the two areas, while also improving traffic conditions.



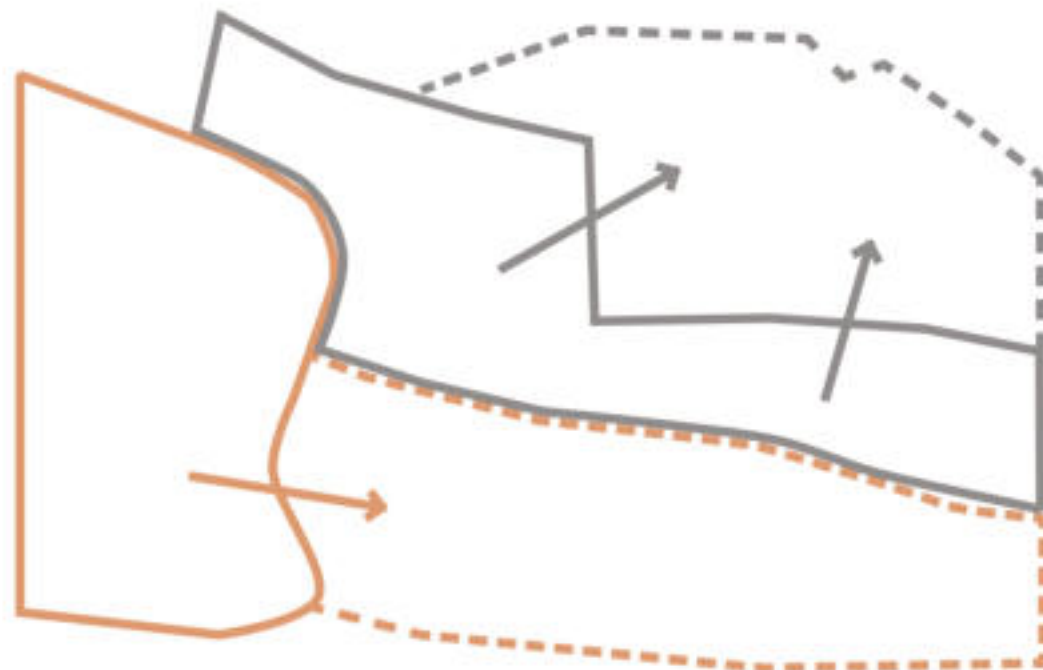
2. GREEN BUFFER ZONE

A green buffer is introduced between the residential area and the industrial zone. The greenery creates a softer separation between the two areas, while improving acoustic and visual insulation from the logistics buildings.



3. ZONING

Possible future growth of the residential area, influenced by its proximity to Prague. As the town continues to expand towards the city, pressure on the industrial zone may increase, forcing logistics development to adapt and potentially shift upwards, towards the airport.



1. MOBILITY



2. GREEN BUFFER ZONE



3. ZONING



DESIGN RESPONSE

The Hostivice industrial zone is understood as a **larger system of warehouses, paved yards, roads, parking areas, technical infrastructure, water systems, and landscape edges**, rather than as a set of isolated buildings.

The project keeps the logistics function of the site, but introduces **green platforms that bridge the residential area and the industrial zone**. These platforms create new connections while softening the boundary between the two areas. Through greenery, public access, photovoltaics, rainwater management, shading, and acoustic buffering, the proposal adds social and environmental value to the logistics landscape.





1. MONOTONE BUILDINGS

The repetitive warehouse volumes create a monotonous built environment and visually disrupt the surrounding landscape.

2. LIMITED ACCESSIBILITY

The industrial zone is mainly organized for trucks and private vehicles, resulting in poor pedestrian accessibility.

3. WEAK TOPOGRAPHY INTEGRATION

The existing development ignores the natural slope of the terrain, missing the opportunity to integrate buildings and landscape more sensitively.

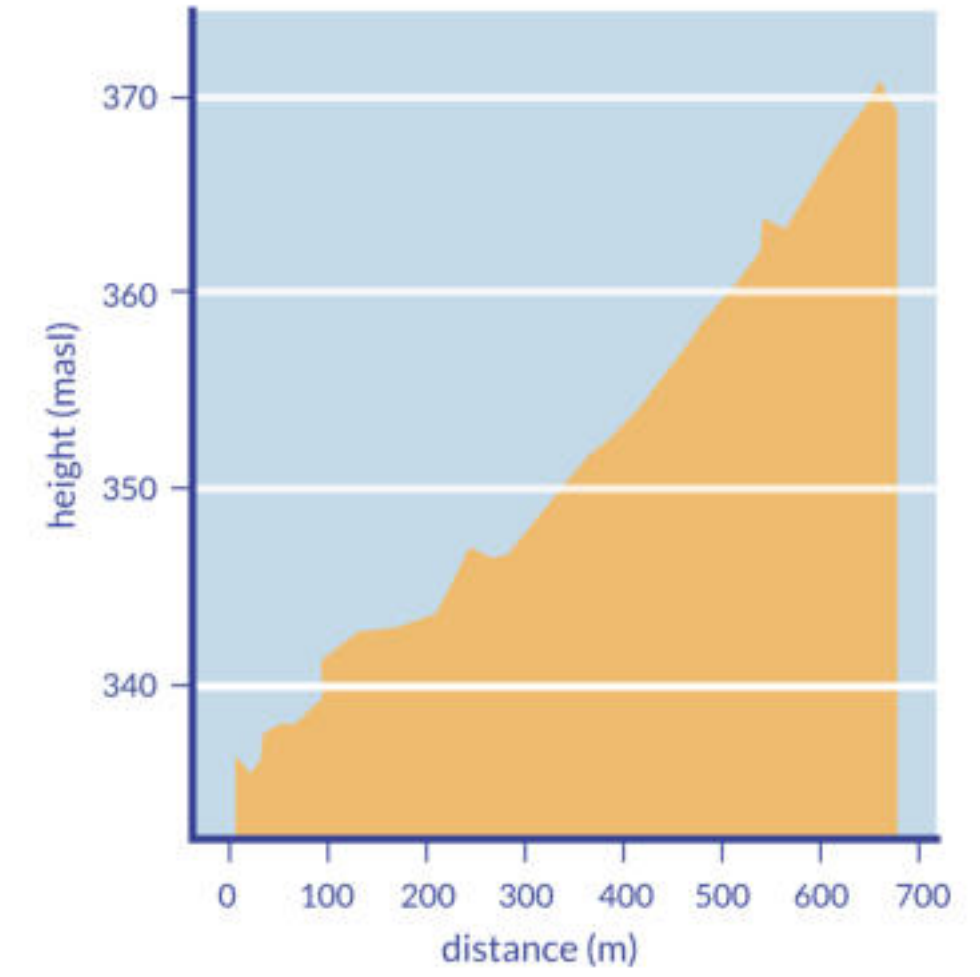
4. INTERRUPTED GREEN-BLUE CONTINUITY

The industrial zone interrupts the continuity of the green-blue corridor, limiting ecological connections between Prague and Hostivice.

INTEREST POINTS

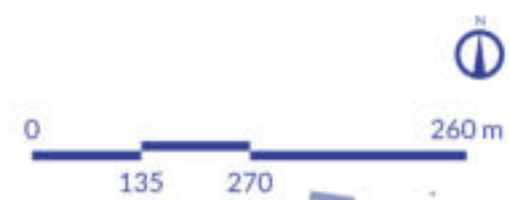
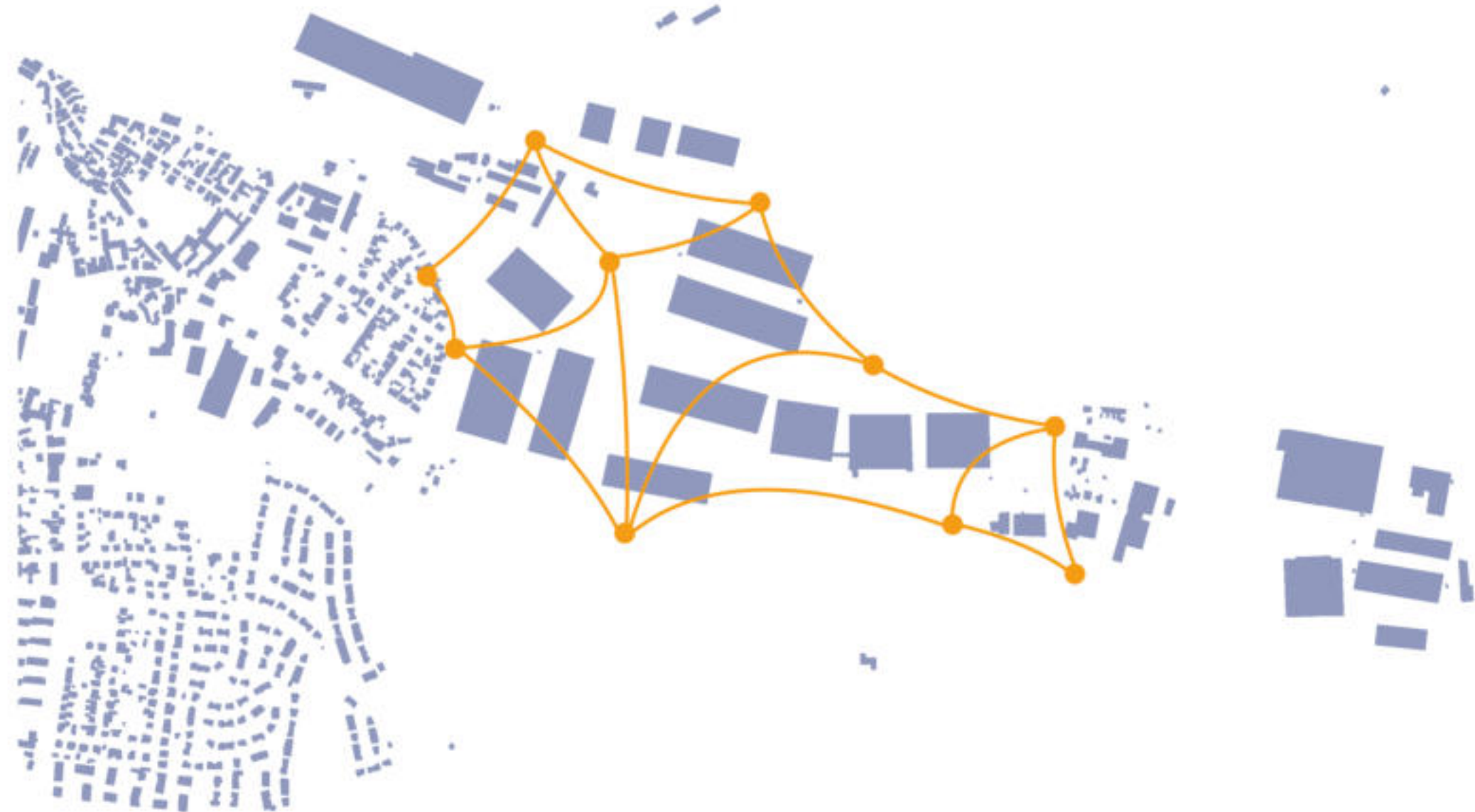


TERRAIN PROFILE

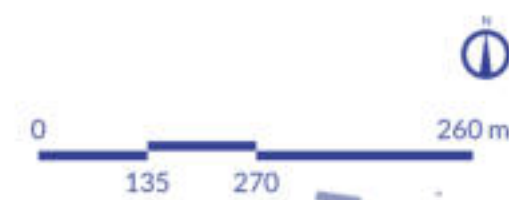


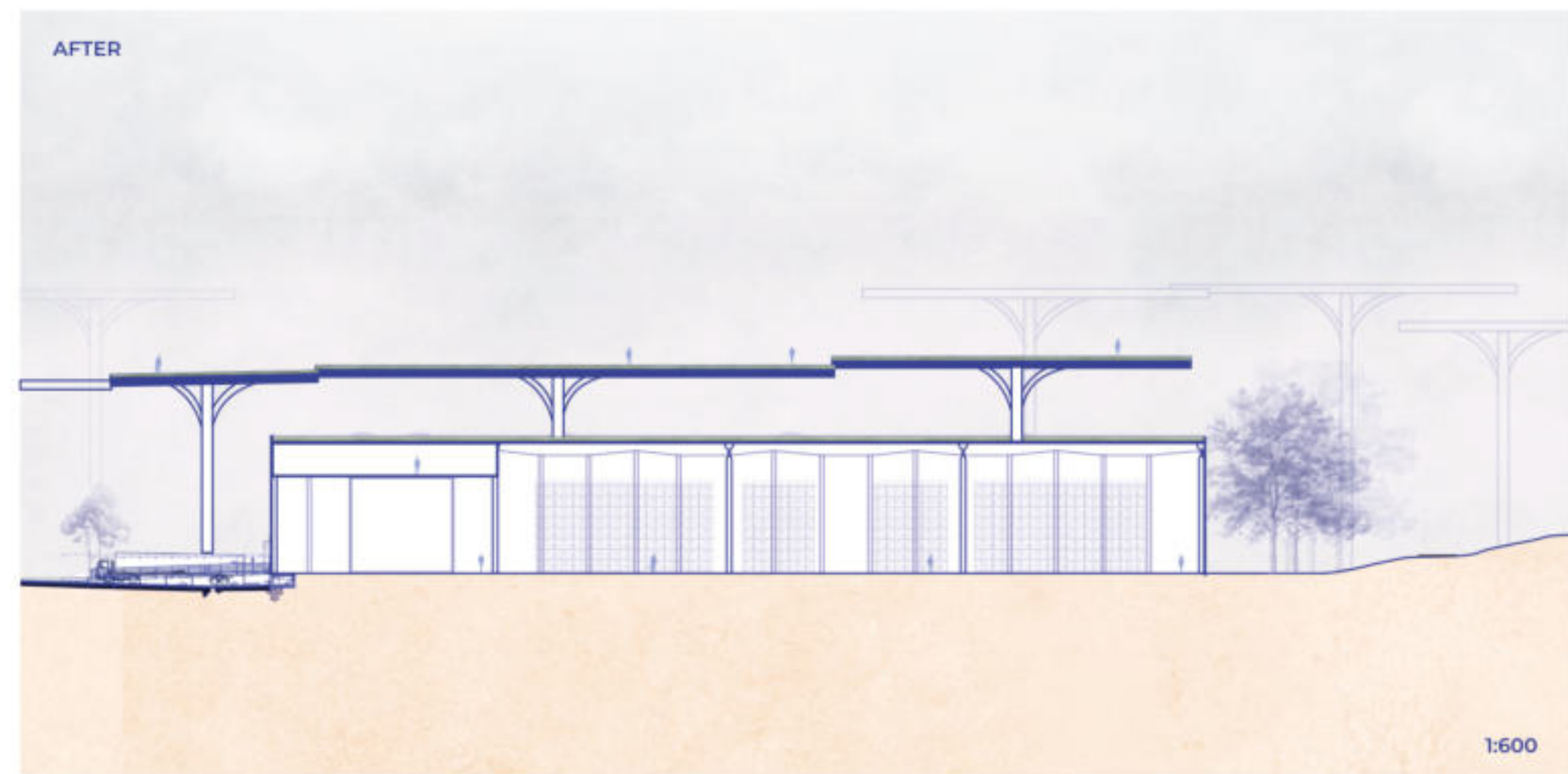
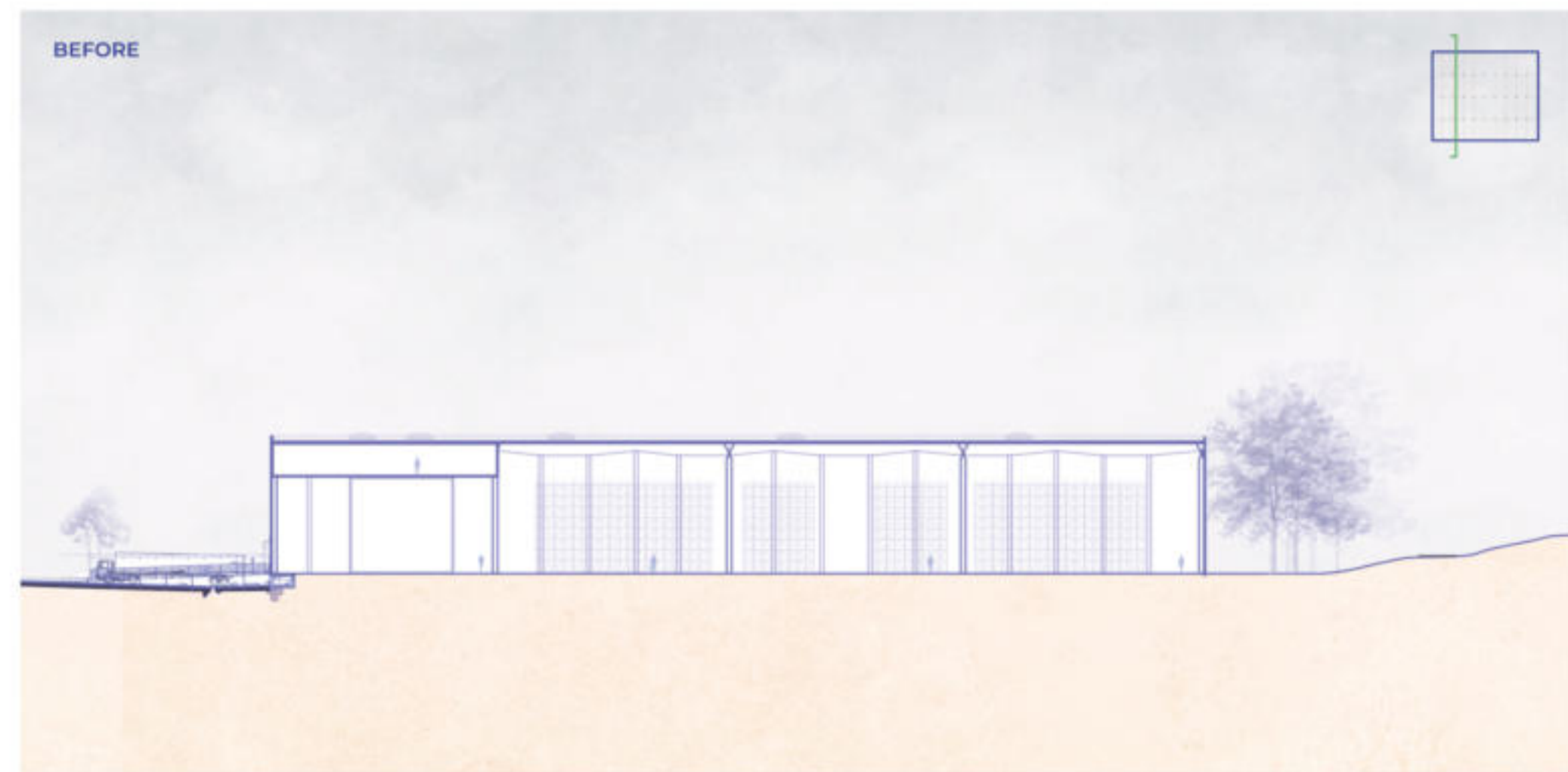
Line length (2D): 677.49 m
Line length (3D): 680.34 m
Tile length: 0.66 m
Number of tiles: 1021
Total descent: 5.42 m
Total ascent: 38.90 m
Max. altitude: 371.03
Min. altitude: 335.48

CONNECTIONS



MEETING POINTS





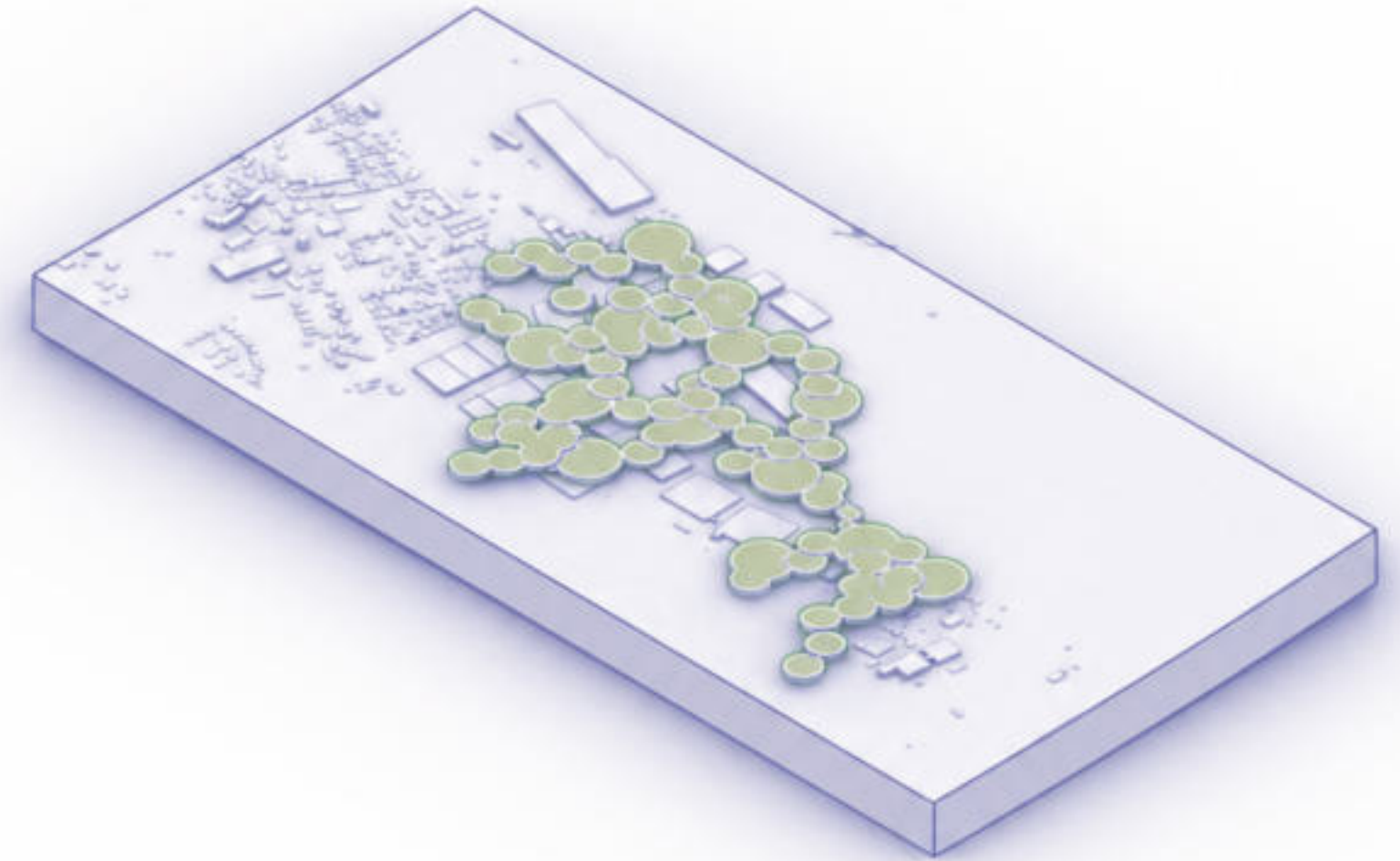


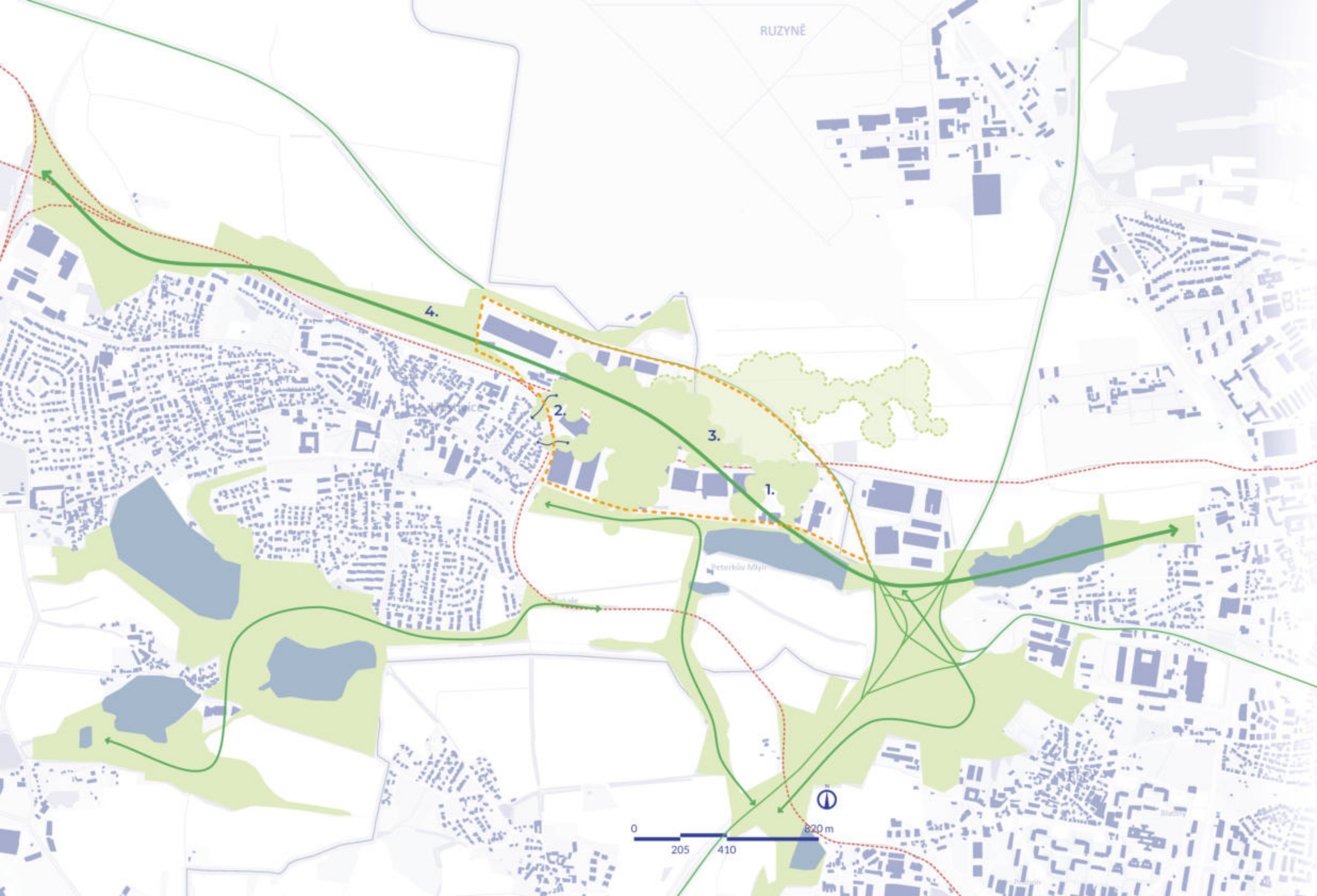
DISCUSSION OF RESULTS

The research shows that logistics zones are efficient as operational systems, but often weak as urban and environmental spaces. In Hostivice, the site benefits from strong regional accessibility, but its local **structure creates barriers, car dependency, limited walkability, sealed surfaces, and pressure on nearby residential areas.**

The case study building confirms that the logistics typology is based on repetition, large spans, prefabrication, and a regular structural grid. These systems support fast construction and efficient storage, but they also rely heavily on concrete-based materials and are usually designed for one main function. This limits adaptability and increases the environmental impact of the building over time.

The project demonstrates that **the scale of logistics architecture can also become an opportunity.** Large roofs can produce **renewable energy** and support **green layers**. Paved surfaces can be reorganized with **water retention, permeable materials, and planting.** Green buffers can **reduce the impact of noise, pollution, and visual disruption on surrounding neighborhoods.**





1. LANDSCAPE-INTEGRATED BUILDINGS

The project breaks down the monotony of the warehouse landscape by adding green roof layers and varied environmental surfaces that help the buildings integrate better with the surroundings.

2. RESIDENTIAL-INDUSTRIAL CONNECTION

Clearer pedestrian and landscape connections between the residential area and the industrial zone.

3. TOPOGRAPHY INTEGRATION

The project responds to the natural slope of the terrain by using layered landscape strategies, water retention areas, and planted transitions that connect buildings more sensitively to the ground.

4. GREEN-BLUE CONTINUITY

The project reconnects the interrupted green-blue corridor, allowing the site to become part of the wider landscape system between Prague and Hostivice.

TABLE OF CONTENTS

- 1. INTRODUCTION
- 2. SITE ANALYSIS
 - 2.1. SITE DEVELOPMENT
 - 2.2. GEOGRAPHIC ANALYSIS
 - 2.3. SOCIAL ANALYSIS
 - 2.4. INFRASTRUCTURE ANALYSIS
 - 2.5. TRANSPORTATION AND SITE TRAFFIC
- 3. STRUCTURE OF THE CASE STUDY BUILDING
 - 3.1. STRUCTURAL SYSTEM
 - 3.2. MECHANICAL PROPERTIES OF STRUCTURAL MATERIALS
 - 3.3. LOADS
- 4. URBAN STRATEGY AND STRUCTURAL TRANSFORMATION
 - 4.1. STRATEGIC SCENARIOS
 - 4.2. PROJECT RESPONSE
- 5. MEP
 - 5.1. ENERGY PRODUCTION
 - 5.2. WATER MANAGEMENT
 - 5.3. HEAT ISLAND, OVERHEATING, AND MICROCLIMATE
 - 5.4. GREEN INFRASTRUCTURE
 - 5.5. AIR QUALITY AND POLLUTION REDUCTION
 - 5.6. ACOUSTIC STRATEGY
 - 5.7. SAFETY, FIRE ACCESS, AND TECHNICAL INFRASTRUCTURE
- 6. SUSTAINABILITY REPORT
 - 6.1. ENVIRONMENTAL SYSTEMS
 - 6.2. MATERIAL AND CONSTRUCTION STRATEGY
 - 6.3. END-OF-LIFE AND CIRCULARITY
- 7. DISCUSSION OF RESULTS
- 8. CONCLUSION
- 9. BIBLIOGRAPHY

1. INTRODUCTION

Logistics buildings are a fundamental part of contemporary urban infrastructure. Their importance has increased with the growth of e-commerce and regional distribution networks. Although these buildings support the everyday functioning of cities, they are usually treated mainly as technical objects, designed for storage capacity, truck access, operational speed, and construction efficiency. Their wider urban, environmental, and social effects are often secondary.

This project focuses on the Hostivice industrial zone on the western edge of Prague. The area has developed as a strategic logistics location because of its proximity to major roads, railway corridors, and Václav Havel Airport Prague. At the same time, it is located close to existing residential neighborhoods, creating a direct relationship between large-scale industrial infrastructure and everyday living environments. This situation raises questions about accessibility, environmental impact, land-use compatibility, and the long-term integration of logistics areas into the urban edge.

The report is based on two related scales of analysis. First, the Hostivice industrial zone is examined as a territorial system shaped by infrastructure, landscape, movement, and its relationship to nearby residential areas. Second, one representative warehouse building is studied as a case study in order to understand the construction logic, structural system, material use, and environmental limitations of the current logistics typology.

Research forms the basis for an urban strategy in which logistics buildings, roof surfaces, paved yards, water systems, and green layers are reconsidered as parts of a larger urban and ecological system.

2. SITE ANALYSIS

Site	Measurements
Site area	approx. 876,000 m²
Developed area / building coverage	approx. 350,000-440,000 m²
Parking area	approx. 35,000 m²

The project site is located in the Hostivice industrial zone on the western edge of Prague. The wider industrial area has an approximate perimeter of 4,700 m and a total surface of around 876,000 m². Its scale shows the importance of the zone within the regional logistics network, but also the strong spatial impact of logistics development on the urban edge. The site is dominated by warehouse halls, loading yards, truck circulation areas, parking surfaces, service zones, and residual open spaces, with limited vegetation and few areas designed for public use.

2.1. SITE DEVELOPMENT

The development of the Hostivice industrial zone has taken place mainly over the past three decades, supported by the area's proximity to major road networks, railway infrastructure, and Václav Havel Airport Prague. These connections have made the area attractive for logistics, storage, and distribution activities.

The zone follows a typical logistics development model. Large warehouse buildings are placed within separate plots and surrounded by loading yards, truck circulation areas, employee and visitor parking, service areas, and technical infrastructure. This layout supports efficient storage and distribution, but it also creates a fragmented urban structure with limited public accessibility and weak connection to the surrounding context.

2.2. GEOGRAPHIC ANALYSIS

The site is located in a relatively flat landscape, which supports large-scale logistics construction and easy vehicle movement. This condition allows wide warehouse footprints, regular structural layouts, and large paved yards to function efficiently. At the same time, it has encouraged extensive land consumption and a high proportion of sealed surfaces.

Large roof areas, asphalt yards, roads, and parking lots reduce the capacity of the ground to absorb rainwater. This increases runoff and creates pressure on drainage systems during heavy rain. The lack of vegetation and shade also contributes to overheating and the urban heat island effect. Nearby ponds, wetlands, and open landscape areas suggest a potential for stronger water management, cooling, and ecological continuity.

2.3. SOCIAL ANALYSIS

The industrial zone is located close to existing residential neighborhoods. This creates a direct tension between logistics operations and everyday living environments. While the area provides employment and economic activity, it also generates noise, truck traffic, air pollution, visual disruption, and reduced spatial quality.

The railway corridor forms a physical boundary between the industrial and residential areas, but it also limits movement and reinforces separation. Public accessibility within the logistics zone is weak. The site is mainly designed for trucks and private cars, while pedestrian and cycling routes are limited, unclear, or secondary. For workers, this reduces comfort and access to outdoor spaces. For residents, the logistics area often appears as a barrier rather than an integrated part of the urban environment.

2.4. INFRASTRUCTURE ANALYSIS

The site is strongly shaped by infrastructure. Its proximity to roads, railway lines, and the airport gives it strong regional accessibility and makes it attractive for logistics development. Goods can move efficiently between Hostivice, Prague, the wider Czech region, and international networks.

However, this infrastructure works differently at the local scale. Roads, railway lines, service routes, and fenced plots divide the area into separate operational zones. Movement is dominated by trucks and private cars. Employee access appears to rely mainly on private car transport, supported by parking areas located close to the warehouse buildings. These parking areas serve staff and visitors and form a large part of the non-building surface of the site.

Truck access is organized through dedicated entrances, internal roads, maneuvering areas, loading docks, and waiting zones. These areas require durable paved surfaces and large turning radii, which further increases the amount of impermeable ground. Public transport access is more limited and often requires walking through spaces not designed primarily for pedestrians. This creates a strong dependence on private vehicles and reinforces the separation between logistics infrastructure and everyday urban movement.

2.5. TRANSPORTATION AND SITE TRAFFIC

Transportation is the main organizing system of the logistics zone. Truck circulation, loading docks, employee parking, service routes, and maneuvering areas determine the layout of the site. Heavy vehicles require durable surfaces, large turning radii, waiting zones, and direct access to warehouse bays.

This traffic logic supports logistics efficiency, but it also produces spatial and environmental problems. Extensive paved areas increase runoff, heat build-up, and the dominance of vehicles. Employee and visitor parking adds another layer of car-based infrastructure, while pedestrian and cycling movement remains secondary. The proximity of residential neighborhoods makes traffic impacts more sensitive, especially noise, emissions, and safety concerns.

3. STRUCTURE OF THE CASE STUDY BUILDING

The selected case-study building is an existing logistics warehouse located within the Hostivice industrial zone. The building represents a typical large-scale logistics typology, with a simple rectangular footprint, a regular structural grid, a flat roof, loading areas, and an office/admin zone inserted within the main warehouse volume.

3.1. STRUCTURAL SYSTEM

The building has an approximate footprint of 115 × 110 m. Based on the plan, the main structural grid is arranged with 11 grid lines in one direction and 5 grid lines in the other direction. This creates 10 bays along the longer side and 4 bays along the shorter side.

Direction	Building length	Number of bays	Approx. spacing
A–K	115 m	10	11.5 m
1–5	110 m	4	27.5 m

3.2. MECHANICAL PROPERTIES OF STRUCTURAL MATERIALS

Material	Use in building	Mechanical property	Density / unit weight
Reinforced concrete C25/30	Columns, foundations, slabs	fck = 25 MPa compression strength	25 kN/m³
Precast / prestressed concrete	Roof girders and office floor panels	High-strength precast concrete	25 kN/m³
Structural steel S355	Bracing, connections, secondary elements	fy = 355 MPa yield strength	78.5 kN/m³

Trapezoidal steel sheet	Roof deck	Structural roof sheeting	0.08 kN/m²
Mineral wool insulation	Roof and wall insulation	Non-structural	0.35 kN/m³
XPS insulation	Ground floor insulation	Non-structural	0.35 kN/m³
Gravel	Sub-base layers	Non-structural fill layer	20 kN/m³
Waterproofing membrane	Roof and floor protection	Non-structural	12 kN/m³

3.3. LOADS

Building part	Dead load	Live load	Snow load
Roof	0.584 kN/m²	0.40 kN/m²	0.56 kN/m²
Office upper floor	10.24 kN/m²	3.00 kN/m²	-
Warehouse ground floor	5.07 kN/m²	5.00 kN/m²	-

4. URBAN STRATEGY AND STRUCTURAL TRANSFORMATION

The urban strategy is based on the site analysis and the case study building. The Hostivice industrial zone is understood as a larger system of warehouses, paved yards, roads, parking areas, technical infrastructure, water systems, and landscape edges, rather than as a set of isolated buildings.

The existing zone works efficiently for logistics, but has limited spatial and environmental quality. Large warehouse volumes are surrounded by asphalt and concrete, green areas are fragmented, and public accessibility is weak. The relationship with nearby residential neighborhoods is mostly defined by separation, traffic, noise, and hard boundaries.

The proposal keeps the logistics function of the site, but adds environmental and social value through green layers and better spatial organization. Roofs are used for photovoltaic panels, green roof systems, and rainwater delay. Paved areas are improved with permeable surfaces, planting, shading, and retention where possible. Residential edges are treated as transition zones with vegetation, acoustic buffers, and landscape elements.

4.1. STRATEGIC SCENARIOS

The urban strategy was developed through three scenarios that explore possible relationships between the residential area, the logistics zone, and the wider landscape context.

The first scenario focuses on mobility. It studies how the residential area and the industrial zone could be better connected through new links and clearer movement systems. The aim is to improve accessibility while also reducing conflicts between local movement and logistics traffic.

The second scenario introduces a green buffer zone between the residential area and the industrial zone. This buffer creates a softer separation between the two areas and improves acoustic and visual insulation from the logistics buildings. It also begins to transform the edge from a hard boundary into a landscape transition.

The third scenario looks at zoning and possible future urban growth. Because Hostivice is close to Prague, residential pressure may increase over time as the town expands toward the city. This could create pressure on the industrial zone and require logistics development to adapt. One possible direction is that logistics functions shift or intensify toward the airport edge, allowing the boundary with the residential area to become more flexible and less conflictual.

4.2. PROJECT RESPONSE

Based on the problems identified through these scenarios, the project proposes a strategy of green layers that connect rather than separate the industrial and residential areas.

The main idea is to bridge the gap between the residential fabric and the industrial zone through a sequence of planted spaces, green buffers, accessible routes, and environmental surfaces. These layers reduce the impact of logistics operations while creating a more gradual transition between everyday urban life and industrial activity.

The strategy also connects the site to the wider green and blue corridor extending from Prague. Existing ponds, wetlands, drainage systems, and open landscape areas are used as a basis for strengthening ecological continuity. In this way, the project works at two scales: locally, it improves the relationship between housing and logistics; territorially, it connects the industrial zone to broader landscape systems.

The result is an urban strategy where logistics remains active, but its edges, roofs, and open surfaces become part of a more integrated environmental framework. The project does not propose a complete replacement of the industrial zone, but a transformation of its structure through landscape, mobility, and ecological systems.

5. MEP

5.1. ENERGY PRODUCTION

The large roof surfaces of the logistics buildings create strong potential for on-site energy production. Photovoltaic panels are placed on suitable warehouse roofs, especially on large uninterrupted surfaces with good solar exposure.

The total site area is approximately 876,000 m². If around 40-50% of the site is covered by buildings, the potential roof area for photovoltaic installation is about 350,000-440,000 m². In Central European conditions, this could produce approximately 56-88 GWh/year, depending on coverage, orientation, and efficiency. This energy can support warehouse lighting, ventilation, heating and cooling systems, offices, technical equipment, and future vehicle charging.

5.2. WATER MANAGEMENT

The site contains many impermeable surfaces, including roofs, asphalt yards, roads, and parking areas. These surfaces increase runoff and place pressure on drainage systems during heavy rain.

The proposal uses a strategy of collecting, slowing, filtering, and reusing rainwater where possible. Water from roofs and paved areas is directed toward retention ponds, planted basins, infiltration zones, and landscape areas. Permeable paving is introduced in areas with lighter traffic, such as parking zones, pedestrian routes, and service spaces. In areas that must remain hard and durable for trucks, water is guided toward planted drainage channels or retention areas.

5.3. HEAT ISLAND, OVERHEATING, AND MICROCLIMATE

The existing logistics zone contributes to overheating because of its large roof surfaces, asphalt yards, parking areas, and limited vegetation. These hard surfaces absorb and store heat, making outdoor working areas, pedestrian routes, and residential edges less comfortable.

The project reduces overheating through lighter or reflective surfaces, shading, vegetation, and water systems. Trees, planted areas, retention ponds, and green roofs help cool the air and reduce surface temperatures. Photovoltaic canopies can also provide shade above parking and loading areas. These strategies improve the microclimate and make the industrial zone more comfortable for workers and less harmful to nearby residential areas.

5.4. GREEN INFRASTRUCTURE

Green infrastructure is used as an environmental system that supports cooling, rainwater management, biodiversity, air quality, acoustic buffering, and spatial organization.

The proposal introduces connected planted areas, tree lines, green edges, retention landscapes, and planting around circulation and parking areas. Roof surfaces also contribute through extensive green layers and combined green roof–photovoltaic systems. These elements help reduce heat build-up, slow rainwater runoff, and create softer transitions between industrial buildings and the surrounding landscape.

5.5. AIR QUALITY AND POLLUTION REDUCTION

Air quality is affected by truck traffic, loading operations, employee cars, dust, and nearby infrastructure. The project responds by organizing movement more clearly and reducing unnecessary driving, waiting, and idling. Heavy truck routes are separated from pedestrian and cycling routes where possible, improving both safety and exposure to pollution.

Renewable energy produced on site can support charging infrastructure for service vehicles, employee cars, and future logistics fleets. Green buffers made of trees and shrubs also help capture dust and filter particles, especially along roads, loading zones, parking areas, and edges facing residential neighborhoods.

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5.6. ACOUSTIC STRATEGY

Noise is produced by truck movement, loading and unloading, reversing signals, technical equipment, and surrounding transport infrastructure.

The acoustic strategy combines spatial organization, façade treatment, and landscape buffers. Noisy functions such as loading bays, truck waiting areas, and technical service zones should be located away from residential edges where possible. Façades facing sensitive areas can use better acoustic insulation and fewer openings. Earth berms, tree belts, dense shrubs, and planted edges reduce the perceived impact of noise and create a softer boundary between the industrial zone and residential areas.

5.7. SAFETY, FIRE ACCESS, AND TECHNICAL INFRASTRUCTURE

Safety is planned at both building and site scale. The strategy uses mainly non-combustible materials in structural and fire-critical areas and supports clear fire separation, evacuation routes, emergency lighting, and signage.

At the site scale, fire vehicles must be able to reach the buildings through clear access routes. Roads, turning spaces, and access points are organized so they are not blocked by parking, loading, or service operations. Roof systems with photovoltaic panels and green layers require safe maintenance access, emergency shut-off systems, and clear electrical organization. Technical infrastructure is therefore integrated into the environmental strategy without reducing safety or operational efficiency.

6. SUSTAINABILITY REPORT

6.1. ENVIRONMENTAL SYSTEMS

The sustainability strategy is considered at the scale of the whole logistics zone. Energy, water, greenery, materials, and circulation are treated as connected systems rather than separate technical topics. The existing site creates several environmental challenges, including sealed surfaces, heat build-up, runoff, pollution, noise, and limited biodiversity. At the same time, its large scale creates opportunities for renewable energy production, rainwater retention, green infrastructure, and improved spatial organization.

6.2. MATERIAL AND CONSTRUCTION STRATEGY

The case study building shows that typical warehouse construction relies heavily on reinforced concrete, precast elements, steel, synthetic insulation, and permanent floor and envelope systems. These materials support durability and fast construction, but they also create high material consumption and embodied carbon.

A more sustainable construction strategy should focus on material efficiency, optimized structural grids, prefabrication, and lower-carbon alternatives. Future logistics buildings could use timber or hybrid structural systems, such as glulam, cross-laminated timber, or combinations of timber with concrete and steel. These systems can reduce carbon impact while maintaining the large spans and durability required for logistics use.

6.3. END-OF-LIFE AND CIRCULARITY

Logistics buildings are often designed as single-purpose structures. When operational needs change, they can be difficult to adapt, and demolition can produce large amounts of waste. The project therefore supports modular structural systems, prefabricated elements, and materials that can be separated, repaired, reused, or replaced.

Clear structural grids and accessible connections make it easier to extend or transform buildings over time. Circularity also applies to the urban scale: roof systems, paved surfaces, technical infrastructure, and landscape layers should be designed to change as logistics, transport, and energy systems evolve. This supports a longer life cycle for the industrial zone and reduces future environmental impact.

7. DISCUSSION OF RESULTS

The research shows that logistics zones are efficient as operational systems, but often weak as urban and environmental spaces. In Hostivice, the site benefits from strong regional accessibility, but its local structure creates barriers, car dependency, limited walkability, sealed surfaces, and pressure on nearby residential areas.

The case study building confirms that the logistics typology is based on repetition, large spans, prefabrication, and a regular structural grid. These systems support fast construction and efficient storage, but they also rely heavily on concrete-based materials and are usually designed for one main function. This limits adaptability and increases the environmental impact of the building over time.

The project demonstrates that the scale of logistics architecture can also become an opportunity. Large roofs can produce renewable energy and support green layers. Paved surfaces can be reorganized with water retention, permeable materials, and planting. Green buffers can reduce the impact of noise, pollution, and visual disruption on surrounding neighborhoods.

8. CONCLUSION

This project proposes a shift in how logistics architecture is understood. Instead of treating warehouses as isolated technical objects, it examines them as parts of a larger urban and environmental system. The Hostivice industrial zone shows how logistics infrastructure can shape the urban edge through transport, land use, material systems, and its relationship with nearby residential areas.

The analysis identifies both the strengths and limitations of the existing typology. The logistics zone is well connected, operationally efficient, and structurally repetitive, but it also creates sealed surfaces, overheating, traffic dominance, weak public accessibility, and high material impact.

The proposed strategy keeps the logistics function of the site, but improves how it performs. Photovoltaic roofs, green layers, water retention, acoustic buffers, clearer circulation, and more adaptable construction principles allow the industrial zone to become more integrated and environmentally responsive. The project suggests that logistics architecture can contribute not only to supply chains, but also to urban resilience, ecological performance, and the quality of the urban edge.

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EPILOGUE

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